Trabalho 3 de Macroeconometria — Anexo de Outputs Brutos

Pass-through cambial e política monetária no Brasil — VAR/VECM

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Maio de 2026

Contents

Este documento é o **anexo de outputs brutos** entregue à professora. Ele reexecuta, linha a linha, o mesmo pipeline econométrico de `trabalho3_econometria.Rmd` — mesmos dados, transformações, parâmetros (Johansen K pelo AIC, VAR p pelo AIC, `lags.pt = 12`, CVs de MacKinnon $k = 3$ no Engle-Granger, ZA modelo `both`) e mesma ordem das seções do paper —, porém imprimindo a saída crua do console R (`print()`, `summary()`, `cat()`) em vez das tabelas formatadas. Nenhum parâmetro foi reotimizado. Toda rotina com bootstrap é precedida de `set.seed()` fixo para reprodutibilidade das bandas.

1. Carga, transformações e estatísticas descritivas

```r
## --- etapa 1: carga e transformações das séries ---
# IPCA - índice histórico IBGE/SIDRA (série 1737)
ipca <- readxl::read_xlsx(file.path(pasta_input, "ipca_sidra_1737.xlsx"))
names(ipca) <- c("data", "mes", "valor")
ipca$data  <- as.Date(ipca$data)
ipca$valor <- as.numeric(ipca$valor)
ipca <- ipca %>%
  mutate(data = as.Date(format(data, "%Y-%m-01"))) %>%
  filter(!is.na(data), !is.na(valor),
         data >= DATA_INICIAL, data <= DATA_FINAL)

# Câmbio R$/US$ - BCB/SGS 3698 (mensal)
cambio_raw <- read.csv2(file.path(pasta_input, "cambio_sgs_3698.csv"),
                        quote = "\"", stringsAsFactors = FALSE)
names(cambio_raw) <- c("data", "valor")
cambio_raw$data  <- as.Date(cambio_raw$data, format = "%d/%m/%Y")
cambio_raw$valor <- as.numeric(gsub(",", ".", cambio_raw$valor))
cambio <- cambio_raw %>% filter(data >= DATA_INICIAL, data <= DATA_FINAL)

# Selic meta - BCB/SGS 432 (diária -> mensal, último valor do mês)
selic_raw <- read.csv2(file.path(pasta_input, "selic_sgs_432.csv"),
                       stringsAsFactors = FALSE)
names(selic_raw) <- c("data", "valor")
selic_raw$data  <- as.Date(selic_raw$data, format = "%d/%m/%Y")
selic_raw$valor <- as.numeric(gsub(",", ".", selic_raw$valor))
selic_raw <- selic_raw %>% filter(!is.na(data), !is.na(valor))
selic_raw$mes <- format(selic_raw$data, "%Y-%m-01")
selic <- selic_raw %>%
  group_by(mes) %>%
  summarise(valor = last(valor), .groups = "drop") %>%
  mutate(data = as.Date(mes)) %>%
  filter(data >= DATA_INICIAL, data <= DATA_FINAL) %>%
  dplyr::select(data, valor)

# IBC-Br dessazonalizado - BCB/SGS 24364 (mensal)
ibcbr_raw <- read.csv2(file.path(pasta_input, "ibcbr_sgs_24364.csv"),
```

```r
                       quote = "\"", stringsAsFactors = FALSE)
names(ibcbr_raw) <- c("data", "valor")
ibcbr_raw$data  <- as.Date(ibcbr_raw$data, format = "%d/%m/%Y")
ibcbr_raw$valor <- as.numeric(gsub(",", ".", ibcbr_raw$valor))
ibcbr <- ibcbr_raw %>% filter(data >= DATA_INICIAL, data <= DATA_FINAL)

# Base alinhada (interseção das datas)
df <- Reduce(function(a, b) merge(a, b, by = "data", all = FALSE),
             list(
               data.frame(data = ipca$data,  ipca_indice = ipca$valor),
               data.frame(data = cambio$data, cambio     = cambio$valor),
               data.frame(data = selic$data,  selic      = selic$valor),
               data.frame(data = ibcbr$data,  ibcbr      = ibcbr$valor)
             ))

# Transformações: log(IPCA), log(câmbio), log(IBC-Br); Selic em nível
df$log_ipca_indice <- log(df$ipca_indice)
df$log_cambio      <- log(df$cambio)
df$log_ibcbr       <- log(df$ibcbr)

cat("Observações:", nrow(df), "| De:", format(min(df$data)),
    "a:", format(max(df$data)), "\n")
```

```
## Observações: 276 | De: 2003-01-01 a: 2025-12-01
```

```r
str(df)
```

```
## 'data.frame':    276 obs. of  8 variables:
##  $ data           : Date, format: "2003-01-01" "2003-02-01" ...
##  $ ipca_indice    : num  2086 2118 2144 2165 2178 ...
##  $ cambio         : num  3.44 3.59 3.45 3.12 2.96 ...
##  $ selic          : num  25.5 26.5 26.5 26.5 26.5 26 24.5 22 20 19 ...
##  $ ibcbr          : num  70.2 71.5 71.1 70.6 69.9 ...
##  $ log_ipca_indice: num  7.64 7.66 7.67 7.68 7.69 ...
##  $ log_cambio     : num  1.24 1.28 1.24 1.14 1.08 ...
##  $ log_ibcbr      : num  4.25 4.27 4.26 4.26 4.25 ...
```

```r
## --- etapa 1: estatísticas descritivas cruas ---
vars_transf <- df[, c("log_ipca_indice", "log_cambio", "selic")]

estat <- data.frame(
  variavel = names(vars_transf),
  n        = sapply(vars_transf, length),
  media    = sapply(vars_transf, mean),
  dp       = sapply(vars_transf, sd),
  min      = sapply(vars_transf, min),
  max      = sapply(vars_transf, max),
  row.names = NULL
)
cat("Estatísticas descritivas das séries transformadas:\n")
```

```
## Estatísticas descritivas das séries transformadas:
```

```r
print(estat)
```

```
##          variavel   n     media        dp       min       max
```

```
## 1 log_ipca_indice 276  8.291574 0.3708595 7.6428502  8.909680
## 2      log_cambio 276  1.105013 0.4130358 0.4471827  1.807797
## 3           selic 276 11.614130 4.6526747 2.0000000 26.500000
```

```r
cat("\nsummary() base das séries transformadas:\n")
```

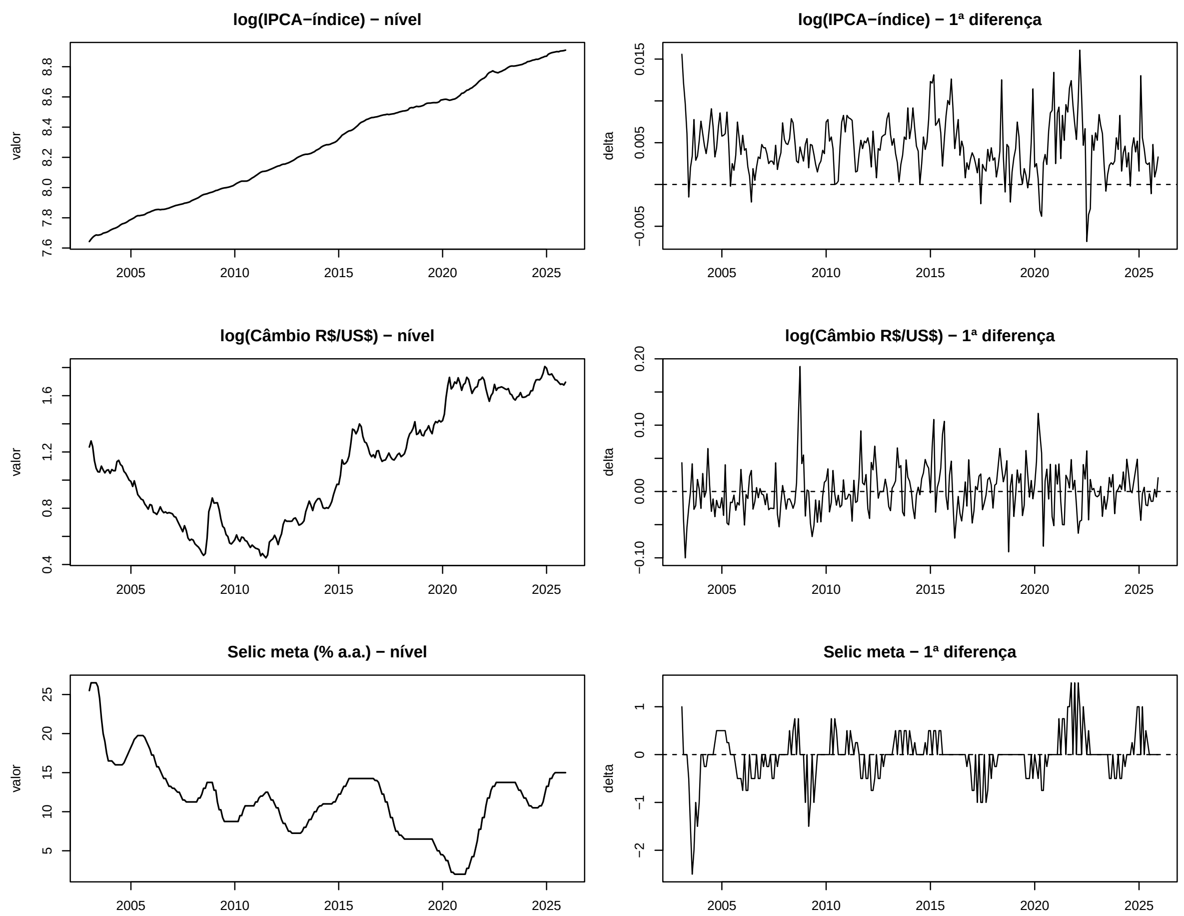
```
##
## summary() base das séries transformadas:
```

```r
print(summary(vars_transf))
```

```
##  log_ipca_indice   log_cambio         selic
##  Min.   :7.643   Min.   :0.4472   Min.   : 2.00
##  1st Qu.:7.962   1st Qu.:0.7519   1st Qu.: 8.75
##  Median :8.284   Median :1.0995   Median :11.50
##  Mean   :8.292   Mean   :1.1050   Mean   :11.61
##  3rd Qu.:8.581   3rd Qu.:1.5621   3rd Qu.:14.25
##  Max.   :8.910   Max.   :1.8078   Max.   :26.50
```

2. Gráficos em nível e 1ª diferença

```r
## --- etapa 2: gráficos em nível e 1ª diferença das 3 séries ---
s_ipca   <- df$log_ipca_indice
s_cambio <- df$log_cambio
s_selic  <- df$selic

par(mfrow = c(3, 2), mar = c(4, 4, 3, 1))

plot(df$data, s_ipca, type = "l", lwd = 1.3,
     main = "log(IPCA-índice) - nível", xlab = "", ylab = "valor")
plot(df$data[-1], diff(s_ipca), type = "l", lwd = 1,
     main = "log(IPCA-índice) - 1ª diferença", xlab = "", ylab = "delta")
abline(h = 0, lty = 2)

plot(df$data, s_cambio, type = "l", lwd = 1.3,
     main = "log(Câmbio R$/US$) - nível", xlab = "", ylab = "valor")
plot(df$data[-1], diff(s_cambio), type = "l", lwd = 1,
     main = "log(Câmbio R$/US$) - 1ª diferença", xlab = "", ylab = "delta")
abline(h = 0, lty = 2)

plot(df$data, s_selic, type = "l", lwd = 1.3,
     main = "Selic meta (% a.a.) - nível", xlab = "", ylab = "valor")
plot(df$data[-1], diff(s_selic), type = "l", lwd = 1,
     main = "Selic meta - 1ª diferença", xlab = "", ylab = "delta")
abline(h = 0, lty = 2)
```



3. Testes de raiz unitária — ADF e KPSS

```
## --- etapa 3: ADF (none/trend/drift, lag por AIC) e KPSS ---
# ADF em NÍVEL - sem constante/tendência (none)
adf_ipca_n <- ur.df(s_ipca,   type = "none", selectlags = "AIC")
adf_cam_n  <- ur.df(s_cambio, type = "none", selectlags = "AIC")
adf_sel_n  <- ur.df(s_selic,  type = "none", selectlags = "AIC")

# ADF em NÍVEL - com constante e tendência (trend)
adf_ipca_n_trend <- ur.df(s_ipca,   type = "trend", selectlags = "AIC")
adf_cam_n_trend  <- ur.df(s_cambio, type = "trend", selectlags = "AIC")
adf_sel_n_trend  <- ur.df(s_selic,  type = "trend", selectlags = "AIC")

# Selic em NÍVEL - com constante (drift), apenas complementar
adf_sel_drift <- ur.df(s_selic, type = "drift", selectlags = "AIC")

# ADF em PRIMEIRA DIFERENÇA - com constante (drift)
adf_ipca_d <- ur.df(diff(s_ipca),   type = "drift", selectlags = "AIC")
adf_cam_d  <- ur.df(diff(s_cambio), type = "drift", selectlags = "AIC")
adf_sel_d  <- ur.df(diff(s_selic),  type = "drift", selectlags = "AIC")

# KPSS (H0: estacionariedade), nível e 1ª diferença
```

```r
kpss_ipca_n <- ur.kpss(s_ipca,   type = "mu")
kpss_cam_n  <- ur.kpss(s_cambio, type = "mu")
kpss_sel_n  <- ur.kpss(s_selic,  type = "mu")
kpss_ipca_d <- ur.kpss(diff(s_ipca),   type = "mu")
kpss_cam_d  <- ur.kpss(diff(s_cambio), type = "mu")
kpss_sel_d  <- ur.kpss(diff(s_selic),  type = "mu")
```

3.1 ADF — saída crua (nível, none)

```r
cat("===== ADF log(IPCA) - nível (none) =====\n");   summary(adf_ipca_n)
```

```
## ===== ADF log(IPCA) - nível (none) =====

##
## ###############################################
## # Augmented Dickey-Fuller Test Unit Root Test #
## ###############################################
##
## Test regression none
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)
##
## Residuals:
##        Min         1Q     Median         3Q        Max
## -0.0126371 -0.0014267 -0.0000194  0.0014983  0.0099691
##
## Coefficients:
##              Estimate Std. Error t value             Pr(>|t|)
## z.lag.1    0.00024445 0.00003373   7.247      0.0000000000044 ***
## z.diff.lag 0.54945503 0.04918856  11.170 < 0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.002729 on 272 degrees of freedom
## Multiple R-squared:  0.7656, Adjusted R-squared:  0.7639
## F-statistic: 444.2 on 2 and 272 DF,  p-value: < 0.00000000000000022
##
##
## Value of test-statistic is: 7.2474
##
## Critical values for test statistics:
##       1pct  5pct 10pct
## tau1 -2.58 -1.95 -1.62
```

```r
cat("\n===== ADF log(Câmbio) - nível (none) =====\n"); summary(adf_cam_n)
```

```
##
## ===== ADF log(Câmbio) - nível (none) =====

##
## ###############################################
## # Augmented Dickey-Fuller Test Unit Root Test #
## ###############################################
```

```
##
## Test regression none
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.107232 -0.020415 -0.000815  0.020152  0.151324
##
## Coefficients:
##             Estimate Std. Error t value     Pr(>|t|)
## z.lag.1    0.0004479  0.0017454   0.257        0.798
## z.diff.lag 0.3358506  0.0570932   5.882 0.0000000118 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.03392 on 272 degrees of freedom
## Multiple R-squared:  0.1142, Adjusted R-squared:  0.1077
## F-statistic: 17.53 on 2 and 272 DF,  p-value: 0.00000006912
##
##
## Value of test-statistic is: 0.2566
##
## Critical values for test statistics:
##       1pct  5pct 10pct
## tau1 -2.58 -1.95 -1.62
```

```
cat("\n===== ADF Selic - nível (none) =====\n");       summary(adf_sel_n)
```

```
##
## ===== ADF Selic - nível (none) =====

##
## ###############################################
## # Augmented Dickey-Fuller Test Unit Root Test #
## ###############################################
##
## Test regression none
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.60141 -0.19146  0.04475  0.28090  1.53851
##
## Coefficients:
##             Estimate Std. Error t value            Pr(>|t|)
## z.lag.1    -0.004163   0.001994  -2.088              0.0377 *
## z.diff.lag  0.531065   0.050288  10.561 <0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 0.4077 on 272 degrees of freedom
## Multiple R-squared:  0.3096, Adjusted R-squared:  0.3046
## F-statistic:    61 on 2 and 272 DF,  p-value: < 0.00000000000000022
##
##
## Value of test-statistic is: -2.0879
##
## Critical values for test statistics:
##       1pct  5pct 10pct
## tau1 -2.58 -1.95 -1.62
```

3.2 ADF — saída crua (nível, trend)

```r
cat("===== ADF log(IPCA) - nível (trend) =====\n");   summary(adf_ipca_n_trend)
```

```
## ===== ADF log(IPCA) - nível (trend) =====
```

```
##
## ###############################################
## # Augmented Dickey-Fuller Test Unit Root Test #
## ###############################################
##
## Test regression trend
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)
##
## Residuals:
##        Min         1Q     Median         3Q        Max
## -0.0118224 -0.0012990 -0.0001148  0.0013751  0.0099914
##
## Coefficients:
##                Estimate  Std. Error t value            Pr(>|t|)
## (Intercept)  0.14067773  0.06002665   2.344              0.0198 *
## z.lag.1     -0.01812400  0.00785009  -2.309              0.0217 *
## tt           0.00008381  0.00003649   2.297              0.0224 *
## z.diff.lag   0.55499080  0.04933659  11.249 <0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.00271 on 270 degrees of freedom
## Multiple R-squared:  0.3232, Adjusted R-squared:  0.3157
## F-statistic: 42.98 on 3 and 270 DF,  p-value: < 0.00000000000000022
##
##
## Value of test-statistic is: -2.3088 19.7487 2.6755
##
## Critical values for test statistics:
##       1pct  5pct 10pct
## tau3 -3.98 -3.42 -3.13
## phi2  6.15  4.71  4.05
## phi3  8.34  6.30  5.36
```

```r
cat("\n===== ADF log(Câmbio) - nível (trend) =====\n"); summary(adf_cam_n_trend)
```

```
##
## ===== ADF log(Câmbio) - nível (trend) =====
##
## ###############################################
## # Augmented Dickey-Fuller Test Unit Root Test #
## ###############################################
##
## Test regression trend
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.106478 -0.019884 -0.001079  0.016433  0.149839
##
## Coefficients:
##                Estimate  Std. Error t value     Pr(>|t|)
## (Intercept)  0.00796256  0.00586609   1.357     0.175792
## z.lag.1     -0.02429806  0.00793962  -3.060     0.002433 **
## tt           0.00014321  0.00004156   3.446     0.000659 ***
## z.diff.lag   0.31930064  0.05640949   5.660 0.0000000385 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.03329 on 270 degrees of freedom
## Multiple R-squared:  0.1514, Adjusted R-squared:  0.142
## F-statistic: 16.06 on 3 and 270 DF,  p-value: 0.000000001231
##
##
## Value of test-statistic is: -3.0604 4.1445 6.097
##
## Critical values for test statistics:
##       1pct  5pct 10pct
## tau3 -3.98 -3.42 -3.13
## phi2  6.15  4.71  4.05
## phi3  8.34  6.30  5.36
```

```r
cat("\n===== ADF Selic - nível (trend) =====\n");       summary(adf_sel_n_trend)
```

```
##
## ===== ADF Selic - nível (trend) =====
##
## ###############################################
## # Augmented Dickey-Fuller Test Unit Root Test #
## ###############################################
##
## Test regression trend
##
##
```

```
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.45590 -0.20166  0.01476  0.22503  1.45859
##
## Coefficients:
##               Estimate Std. Error t value             Pr(>|t|)
## (Intercept)  0.1282078  0.1079345   1.188              0.23594
## z.lag.1     -0.0164489  0.0061232  -2.686              0.00767 **
## tt           0.0002854  0.0003621   0.788              0.43132
## z.diff.lag   0.5142022  0.0507784  10.126 < 0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4023 on 270 degrees of freedom
## Multiple R-squared:  0.3279, Adjusted R-squared:  0.3204
## F-statistic:  43.9 on 3 and 270 DF,  p-value: < 0.00000000000000022
##
##
## Value of test-statistic is: -2.6863 4.619 6.5461
##
## Critical values for test statistics:
##       1pct  5pct 10pct
## tau3 -3.98 -3.42 -3.13
## phi2  6.15  4.71  4.05
## phi3  8.34  6.30  5.36
```

```r
cat("\n===== ADF Selic - nível (drift; complementar) =====\n"); summary(adf_sel_drift)
```

```
##
## ===== ADF Selic - nível (drift; complementar) =====
```

```
##
## ###############################################
## # Augmented Dickey-Fuller Test Unit Root Test #
## ###############################################
##
## Test regression drift
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.45548 -0.20061  0.00923  0.22511  1.47864
##
## Coefficients:
##              Estimate Std. Error t value             Pr(>|t|)
## (Intercept)  0.195483   0.066009   2.961             0.003333 **
## z.lag.1     -0.018824   0.005327  -3.534             0.000481 ***
## z.diff.lag   0.522384   0.049671  10.517 < 0.0000000000000002 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.402 on 271 degrees of freedom
## Multiple R-squared:  0.3263, Adjusted R-squared:  0.3213
## F-statistic: 65.63 on 2 and 271 DF,  p-value: < 0.00000000000000022
##
##
## Value of test-statistic is: -3.5339 6.6272
##
## Critical values for test statistics:
##       1pct  5pct 10pct
## tau2 -3.44 -2.87 -2.57
## phi1  6.47  4.61  3.79
```

3.3 ADF — saída crua (1ª diferença, drift)

```
cat("===== ADF log(IPCA) - 1ª dif. (drift) =====\n");   summary(adf_ipca_d)
```

```
## ===== ADF log(IPCA) - 1ª dif. (drift) =====

##
## ###############################################
## # Augmented Dickey-Fuller Test Unit Root Test #
## ###############################################
##
## Test regression drift
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)
##
## Residuals:
##        Min         1Q     Median         3Q        Max
## -0.0124555 -0.0014312 -0.0000663  0.0014531  0.0099951
##
## Coefficients:
##               Estimate Std. Error t value         Pr(>|t|)
## (Intercept)  0.0019964  0.0003087   6.467 0.00000000046657 ***
## z.lag.1     -0.4442931  0.0567203  -7.833 0.00000000000011 ***
## z.diff.lag  -0.0378421  0.0596315  -0.635            0.526
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.002732 on 270 degrees of freedom
## Multiple R-squared:  0.2365, Adjusted R-squared:  0.2308
## F-statistic: 41.81 on 2 and 270 DF,  p-value: < 0.00000000000000022
##
##
## Value of test-statistic is: -7.8331 30.716
##
## Critical values for test statistics:
##       1pct  5pct 10pct
## tau2 -3.44 -2.87 -2.57
## phi1  6.47  4.61  3.79
```

```r
cat("\n===== ADF log(Câmbio) - 1ª dif. (drift) =====\n"); summary(adf_cam_d)
```

```
##
## ===== ADF log(Câmbio) - 1ª dif. (drift) =====
##
## ###############################################
## # Augmented Dickey-Fuller Test Unit Root Test #
## ###############################################
##
## Test regression drift
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.107331 -0.021145 -0.001864  0.019388  0.147624
##
## Coefficients:
##              Estimate Std. Error t value            Pr(>|t|)
## (Intercept)  0.001277   0.002045   0.624               0.533
## z.lag.1     -0.711781   0.069453 -10.248 <0.0000000000000002 ***
## z.diff.lag   0.081793   0.060247   1.358               0.176
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.03374 on 270 degrees of freedom
## Multiple R-squared:  0.3344, Adjusted R-squared:  0.3294
## F-statistic: 67.82 on 2 and 270 DF,  p-value: < 0.00000000000000022
##
##
## Value of test-statistic is: -10.2484 52.5181
##
## Critical values for test statistics:
##       1pct  5pct 10pct
## tau2 -3.44 -2.87 -2.57
## phi1  6.47  4.61  3.79
```

```r
cat("\n===== ADF Selic - 1ª dif. (drift) =====\n");       summary(adf_sel_d)
```

```
##
## ===== ADF Selic - 1ª dif. (drift) =====
##
## ###############################################
## # Augmented Dickey-Fuller Test Unit Root Test #
## ###############################################
##
## Test regression drift
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.78815 -0.23677  0.01323  0.19296  0.94356
##
## Coefficients:
##             Estimate Std. Error t value        Pr(>|t|)
## (Intercept) -0.01323    0.02305  -0.574           0.566
## z.lag.1     -0.28107    0.05332  -5.271 0.0000002775515 ***
## z.diff.lag  -0.37978    0.05556  -6.835 0.0000000000544 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3792 on 270 degrees of freedom
## Multiple R-squared:  0.3405, Adjusted R-squared:  0.3356
## F-statistic:  69.7 on 2 and 270 DF,  p-value: < 0.00000000000000022
##
##
## Value of test-statistic is: -5.2714 13.897
##
## Critical values for test statistics:
##       1pct  5pct 10pct
## tau2 -3.44 -2.87 -2.57
## phi1  6.47  4.61  3.79
```

3.4 KPSS — saída crua

```r
cat("===== KPSS log(IPCA) - nível =====\n");   summary(kpss_ipca_n)
```

```
## ===== KPSS log(IPCA) - nível =====

##
## #######################
## # KPSS Unit Root Test #
## #######################
##
## Test is of type: mu with 5 lags.
##
## Value of test-statistic is: 4.7031
##
## Critical value for a significance level of:
##                 10pct  5pct 2.5pct  1pct
## critical values 0.347 0.463  0.574 0.739
```

```r
cat("\n===== KPSS log(Câmbio) - nível =====\n"); summary(kpss_cam_n)
```

```
##
## ===== KPSS log(Câmbio) - nível =====

##
## #######################
## # KPSS Unit Root Test #
## #######################
##
## Test is of type: mu with 5 lags.
##
```

```
## Value of test-statistic is: 3.4759
##
## Critical value for a significance level of:
##                 10pct  5pct 2.5pct  1pct
## critical values 0.347 0.463  0.574 0.739
```

```r
cat("\n===== KPSS Selic - nível =====\n");        summary(kpss_sel_n)
```

```
##
## ===== KPSS Selic - nível =====
##
## #######################
## # KPSS Unit Root Test #
## #######################
##
## Test is of type: mu with 5 lags.
##
## Value of test-statistic is: 1.5567
##
## Critical value for a significance level of:
##                 10pct  5pct 2.5pct  1pct
## critical values 0.347 0.463  0.574 0.739
```

```r
cat("\n===== KPSS log(IPCA) - 1ª dif. =====\n"); summary(kpss_ipca_d)
```

```
##
## ===== KPSS log(IPCA) - 1ª dif. =====
##
## #######################
## # KPSS Unit Root Test #
## #######################
##
## Test is of type: mu with 5 lags.
##
## Value of test-statistic is: 0.0895
##
## Critical value for a significance level of:
##                 10pct  5pct 2.5pct  1pct
## critical values 0.347 0.463  0.574 0.739
```

```r
cat("\n===== KPSS log(Câmbio) - 1ª dif. =====\n"); summary(kpss_cam_d)
```

```
##
## ===== KPSS log(Câmbio) - 1ª dif. =====
##
## #######################
## # KPSS Unit Root Test #
## #######################
##
## Test is of type: mu with 5 lags.
##
## Value of test-statistic is: 0.4019
##
## Critical value for a significance level of:
##                 10pct  5pct 2.5pct  1pct
```

```
## critical values 0.347 0.463  0.574 0.739
```

```r
cat("\n===== KPSS Selic - 1ª dif. =====\n");     summary(kpss_sel_d)
```

```
##
## ===== KPSS Selic - 1ª dif. =====
##
## #######################
## # KPSS Unit Root Test #
## #######################
##
## Test is of type: mu with 5 lags.
##
## Value of test-statistic is: 0.3938
##
## Critical value for a significance level of:
##                 10pct  5pct 2.5pct  1pct
## critical values 0.347 0.463  0.574 0.739
```

4. Zivot-Andrews (quebra estrutural endógena)

```r
## --- etapa 4: Zivot-Andrews modelo both, lag NULL e lag 1 ---
# NÍVEL - lag NULL
za_ipca_n <- ur.za(ts(s_ipca),   model = "both", lag = NULL)
za_cam_n  <- ur.za(ts(s_cambio), model = "both", lag = NULL)
za_sel_n  <- ur.za(ts(s_selic),  model = "both", lag = NULL)
# NÍVEL - lag 1
za_ipca_n_1 <- ur.za(ts(s_ipca),   model = "both", lag = 1)
za_cam_n_1  <- ur.za(ts(s_cambio), model = "both", lag = 1)
za_sel_n_1  <- ur.za(ts(s_selic),  model = "both", lag = 1)
# 1ª DIFERENÇA - lag NULL
za_ipca_d <- ur.za(ts(diff(s_ipca)),   model = "both", lag = NULL)
za_cam_d  <- ur.za(ts(diff(s_cambio)), model = "both", lag = NULL)
za_sel_d  <- ur.za(ts(diff(s_selic)),  model = "both", lag = NULL)
# 1ª DIFERENÇA - lag 1
za_ipca_d_1 <- ur.za(ts(diff(s_ipca)),   model = "both", lag = 1)
za_cam_d_1  <- ur.za(ts(diff(s_cambio)), model = "both", lag = 1)
za_sel_d_1  <- ur.za(ts(diff(s_selic)),  model = "both", lag = 1)

cat("CV 5% tabelado (Zivot & Andrews, 1992; modelo 'both'): -5.08\n")
```

```
## CV 5% tabelado (Zivot & Andrews, 1992; modelo 'both'): -5.08
```

```r
cat("Datas amostrais para mapear o ponto de quebra (@bpoint):\n")
```

```
## Datas amostrais para mapear o ponto de quebra (@bpoint):
```

```r
cat("  nível: df$data[1] =", format(df$data[1]),
    "... df$data[", nrow(df), "] =", format(df$data[nrow(df)]), "\n")
```

```
##   nível: df$data[1] = 2003-01-01 ... df$data[ 276 ] = 2025-12-01
```

4.1 ZA — nível (lag NULL e lag 1)

```
cat("===== ZA log(IPCA) nível (lag NULL) =====\n");   summary(za_ipca_n)
```

```
## ===== ZA log(IPCA) nível (lag NULL) =====
##
## ################################
## # Zivot-Andrews Unit Root Test #
## ################################
##
##
## Call:
## lm(formula = testmat)
##
## Residuals:
##        Min         1Q     Median         3Q        Max
## -0.0099043 -0.0020648 -0.0001453  0.0017957  0.0123271
##
## Coefficients:
##                Estimate  Std. Error t value             Pr(>|t|)
## (Intercept)  0.23038758  0.09047986   2.546              0.01144 *
## y.l1         0.97055094  0.01181419  82.151 < 0.0000000000000002 ***
## trend        0.00012547  0.00005198   2.414              0.01646 *
## du           0.00261111  0.00098722   2.645              0.00865 **
## dt          -0.00001146  0.00001005  -1.140              0.25542
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.003298 on 270 degrees of freedom
##   (1 observation deleted due to missingness)
## Multiple R-squared:  0.9999, Adjusted R-squared:  0.9999
## F-statistic: 8.598e+05 on 4 and 270 DF,  p-value: < 0.00000000000000022
##
##
## Teststatistic: -2.4927
## Critical values: 0.01= -5.57 0.05= -5.08 0.1= -4.82
##
## Potential break point at position: 143
```

```
cat("\n===== ZA log(Câmbio) nível (lag NULL) =====\n"); summary(za_cam_n)
```

```
##
## ===== ZA log(Câmbio) nível (lag NULL) =====
##
## ################################
## # Zivot-Andrews Unit Root Test #
## ################################
##
##
## Call:
## lm(formula = testmat)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
```

```
## -0.092122 -0.021251 -0.002381  0.016906  0.186283
##
## Coefficients:
##               Estimate Std. Error t value           Pr(>|t|)
## (Intercept)  0.0003227  0.0228307   0.014             0.9887
## y.l1         0.9665742  0.0101626  95.111 <0.0000000000000002 ***
## trend        0.0031182  0.0017353   1.797             0.0735 .
## du          -0.0461010  0.0177992  -2.590             0.0101 *
## dt          -0.0028969  0.0017390  -1.666             0.0969 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.035 on 270 degrees of freedom
##   (1 observation deleted due to missingness)
## Multiple R-squared:  0.9929, Adjusted R-squared:  0.9928
## F-statistic:  9503 on 4 and 270 DF,  p-value: < 0.00000000000000022
##
##
## Teststatistic: -3.2891
## Critical values: 0.01= -5.57 0.05= -5.08 0.1= -4.82
##
## Potential break point at position: 18
```

```
cat("\n===== ZA Selic nível (lag NULL) =====\n");       summary(za_sel_n)
```

```
##
## ===== ZA Selic nível (lag NULL) =====

##
## ################################
## # Zivot-Andrews Unit Root Test #
## ################################
##
##
## Call:
## lm(formula = testmat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.03044 -0.20106  0.00345  0.20660  1.50327
##
## Coefficients:
##               Estimate Std. Error t value             Pr(>|t|)
## (Intercept)  0.2638672  0.1692007   1.559             0.120051
## y.l1         0.9698696  0.0086960 111.531 < 0.0000000000000002 ***
## trend        0.0005956  0.0008371   0.712             0.477369
## du          -0.4577781  0.1240581  -3.690             0.000271 ***
## dt           0.0069923  0.0021066   3.319             0.001027 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4659 on 270 degrees of freedom
##   (1 observation deleted due to missingness)
## Multiple R-squared:  0.9898, Adjusted R-squared:  0.9897
## F-statistic:  6565 on 4 and 270 DF,  p-value: < 0.00000000000000022
```

```
##
##
## Teststatistic: -3.4649
## Critical values: 0.01= -5.57 0.05= -5.08 0.1= -4.82
##
## Potential break point at position: 173
```

```
cat("\n===== ZA log(IPCA) nível (lag 1) =====\n");      summary(za_ipca_n_1)
```

```
##
## ===== ZA log(IPCA) nível (lag 1) =====
##
## ################################
## # Zivot-Andrews Unit Root Test #
## ################################
##
##
## Call:
## lm(formula = testmat)
##
## Residuals:
##        Min         1Q     Median         3Q        Max
## -0.0111878 -0.0013764 -0.0000532  0.0013634  0.0104245
##
## Coefficients:
##                 Estimate   Std. Error t value            Pr(>|t|)
## (Intercept)  0.239992484  0.073619996   3.260             0.00126 **
## y.l1         0.968912917  0.009613087 100.791 < 0.0000000000000002 ***
## trend        0.000137078  0.000042211   3.247             0.00131 **
## y.dl1        0.547324471  0.049080097  11.152 < 0.0000000000000002 ***
## du           0.001925042  0.000803151   2.397             0.01722 *
## dt          -0.000007555  0.000008250  -0.916             0.36066
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.002687 on 268 degrees of freedom
##   (2 observations deleted due to missingness)
## Multiple R-squared:  0.9999, Adjusted R-squared:  0.9999
## F-statistic: 1.025e+06 on 5 and 268 DF,  p-value: < 0.00000000000000022
##
##
## Teststatistic: -3.2338
## Critical values: 0.01= -5.57 0.05= -5.08 0.1= -4.82
##
## Potential break point at position: 142
```

```
cat("\n===== ZA log(Câmbio) nível (lag 1) =====\n");    summary(za_cam_n_1)
```

```
##
## ===== ZA log(Câmbio) nível (lag 1) =====
##
## ################################
## # Zivot-Andrews Unit Root Test #
## ################################
```

```
##
##
## Call:
## lm(formula = testmat)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.104012 -0.019454 -0.004314  0.014997  0.148791
##
## Coefficients:
##                Estimate Std. Error t value             Pr(>|t|)
## (Intercept)  0.0635555  0.0204248   3.112              0.00206 **
## y.l1         0.9327582  0.0173471  53.770 < 0.0000000000000002 ***
## trend       -0.0003061  0.0001483  -2.064              0.03999 *
## y.dl1        0.3374699  0.0570673   5.914         0.0000000102 ***
## du           0.0307956  0.0098002   3.142              0.00186 **
## dt           0.0007037  0.0002552   2.758              0.00622 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.03275 on 268 degrees of freedom
##   (2 observations deleted due to missingness)
## Multiple R-squared:  0.9939, Adjusted R-squared:  0.9938
## F-statistic:  8684 on 5 and 268 DF,  p-value: < 0.00000000000000022
##
##
## Teststatistic: -3.8762
## Critical values: 0.01= -5.57 0.05= -5.08 0.1= -4.82
##
## Potential break point at position: 110
```

```r
cat("\n===== ZA Selic nível (lag 1) =====\n");          summary(za_sel_n_1)
```

```
##
## ===== ZA Selic nível (lag 1) =====

##
## ################################
## # Zivot-Andrews Unit Root Test #
## ################################
##
##
## Call:
## lm(formula = testmat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.46587 -0.20231 -0.00085  0.21689  1.46649
##
## Coefficients:
##                Estimate Std. Error t value             Pr(>|t|)
## (Intercept)  0.3654193  0.1439687   2.538              0.01171 *
## y.l1         0.9695056  0.0074699 129.788 < 0.0000000000000002 ***
## trend       -0.0001432  0.0007155  -0.200              0.84148
## y.dl1        0.4863565  0.0505860   9.614 < 0.0000000000000002 ***
```

```
## du          -0.3255262  0.1071715  -3.037             0.00262 **
## dt           0.0054147  0.0017986   3.011             0.00286 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.395 on 268 degrees of freedom
##   (2 observations deleted due to missingness)
## Multiple R-squared:  0.9924, Adjusted R-squared:  0.9923
## F-statistic:  7041 on 5 and 268 DF,  p-value: < 0.00000000000000022
##
##
## Teststatistic: -4.0823
## Critical values: 0.01= -5.57 0.05= -5.08 0.1= -4.82
##
## Potential break point at position: 173
```

4.2 ZA — 1ª diferença (lag NULL e lag 1)

```r
cat("===== ZA log(IPCA) 1ª dif. (lag NULL) =====\n");   summary(za_ipca_d)
```

```
## ===== ZA log(IPCA) 1ª dif. (lag NULL) =====

##
## ################################
## # Zivot-Andrews Unit Root Test #
## ################################
##
##
## Call:
## lm(formula = testmat)
##
## Residuals:
##        Min         1Q     Median         3Q        Max
## -0.0131876 -0.0013075 -0.0000279  0.0014849  0.0107931
##
## Coefficients:
##                 Estimate   Std. Error t value             Pr(>|t|)
## (Intercept)  0.002423056  0.000462698   5.237           0.00000033 ***
## y.l1         0.504921286  0.050884489   9.923 < 0.0000000000000002 ***
## trend       -0.000002511  0.000003096  -0.811              0.41808
## du           0.002269163  0.000806769   2.813              0.00528 **
## dt          -0.000048362  0.000018984  -2.548              0.01141 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.002699 on 269 degrees of freedom
##   (1 observation deleted due to missingness)
## Multiple R-squared:  0.3311, Adjusted R-squared:  0.3212
## F-statistic: 33.29 on 4 and 269 DF,  p-value: < 0.00000000000000022
##
##
## Teststatistic: -9.7295
## Critical values: 0.01= -5.57 0.05= -5.08 0.1= -4.82
##
```

```
## Potential break point at position: 211
```

```r
cat("\n===== ZA log(Câmbio) 1ª dif. (lag NULL) =====\n"); summary(za_cam_d)
```

```
##
## ===== ZA log(Câmbio) 1ª dif. (lag NULL) =====
##
## ################################
## # Zivot-Andrews Unit Root Test #
## ################################
##
##
## Call:
## lm(formula = testmat)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.105552 -0.020289 -0.001054  0.017536  0.157475
##
## Coefficients:
##                Estimate  Std. Error t value    Pr(>|t|)
## (Intercept) -0.01583646  0.00562304  -2.816     0.00522 **
## y.l1         0.29304591  0.05763506   5.085 0.000000691 ***
## trend        0.00021237  0.00006395   3.321     0.00102 **
## du          -0.01664722  0.00816259  -2.039     0.04238 *
## dt          -0.00018142  0.00010643  -1.705     0.08942 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.03341 on 269 degrees of freedom
##   (1 observation deleted due to missingness)
## Multiple R-squared:  0.1487, Adjusted R-squared:  0.136
## F-statistic: 11.74 on 4 and 269 DF,  p-value: 0.000000008317
##
##
## Teststatistic: -12.266
## Critical values: 0.01= -5.57 0.05= -5.08 0.1= -4.82
##
## Potential break point at position: 152
```

```r
cat("\n===== ZA Selic 1ª dif. (lag NULL) =====\n");       summary(za_sel_d)
```

```
##
## ===== ZA Selic 1ª dif. (lag NULL) =====
##
## ################################
## # Zivot-Andrews Unit Root Test #
## ################################
##
##
## Call:
## lm(formula = testmat)
##
## Residuals:
```

```
##      Min       1Q   Median       3Q      Max
## -1.45736 -0.21789 -0.00407  0.20800  1.46279
##
## Coefficients:
##             Estimate Std. Error t value            Pr(>|t|)
## (Intercept)  0.42063    0.41219   1.020             0.30842
## y.l1         0.44147    0.05271   8.375 0.00000000000000309 ***
## trend       -0.22569    0.07855  -2.873             0.00439 **
## du           1.30783    0.28737   4.551 0.00000809320262047 ***
## dt           0.22622    0.07857   2.879             0.00431 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3915 on 269 degrees of freedom
##   (1 observation deleted due to missingness)
## Multiple R-squared:  0.3657, Adjusted R-squared:  0.3563
## F-statistic: 38.77 on 4 and 269 DF,  p-value: < 0.0000000000000022
##
##
## Teststatistic: -10.5953
## Critical values: 0.01= -5.57 0.05= -5.08 0.1= -4.82
##
## Potential break point at position: 8
```

```
cat("\n===== ZA log(IPCA) 1ª dif. (lag 1) =====\n");      summary(za_ipca_d_1)
```

```
##
## ===== ZA log(IPCA) 1ª dif. (lag 1) =====
##
## ################################
## # Zivot-Andrews Unit Root Test #
## ################################
##
##
## Call:
## lm(formula = testmat)
##
## Residuals:
##        Min         1Q     Median         3Q        Max
## -0.0131569 -0.0013144 -0.0000058  0.0014790  0.0107395
##
## Coefficients:
##                 Estimate   Std. Error t value             Pr(>|t|)
## (Intercept)  0.002383236  0.000488691   4.877 0.000001854287269220 ***
## y.l1         0.504647501  0.058980422   8.556 0.000000000000000928 ***
## trend       -0.000002223  0.000003129  -0.710              0.47810
## y.dl1       -0.015603718  0.059629274  -0.262              0.79377
## du           0.002253398  0.000816401   2.760              0.00618 **
## dt          -0.000048783  0.000019159  -2.546              0.01145 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.002706 on 267 degrees of freedom
##   (2 observations deleted due to missingness)
```

```
## Multiple R-squared:  0.319,  Adjusted R-squared:  0.3062
## F-statistic: 25.01 on 5 and 267 DF,  p-value: < 0.00000000000000022
##
##
## Teststatistic: -8.3986
## Critical values: 0.01= -5.57 0.05= -5.08 0.1= -4.82
##
## Potential break point at position: 211
```

```
cat("\n===== ZA log(Câmbio) 1ª dif. (lag 1) =====\n");    summary(za_cam_d_1)
```

```
##
## ===== ZA log(Câmbio) 1ª dif. (lag 1) =====
```

```
##
## ################################
## # Zivot-Andrews Unit Root Test #
## ################################
##
##
## Call:
## lm(formula = testmat)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.104516 -0.019701 -0.002838  0.017230  0.154588
##
## Coefficients:
##                Estimate  Std. Error t value Pr(>|t|)
## (Intercept) -0.01542781  0.00567391  -2.719  0.00698 **
## y.l1         0.21913728  0.07163634   3.059  0.00245 **
## trend        0.00020611  0.00006305   3.269  0.00122 **
## y.dl1        0.11391547  0.06019594   1.892  0.05952 .
## du          -0.01683679  0.00823225  -2.045  0.04181 *
## dt          -0.00017201  0.00010877  -1.581  0.11496
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.03326 on 267 degrees of freedom
##   (2 observations deleted due to missingness)
## Multiple R-squared:  0.158,  Adjusted R-squared:  0.1422
## F-statistic: 10.02 on 5 and 267 DF,  p-value: 0.000000008427
##
##
## Teststatistic: -10.9004
## Critical values: 0.01= -5.57 0.05= -5.08 0.1= -4.82
##
## Potential break point at position: 156
```

```
cat("\n===== ZA Selic 1ª dif. (lag 1) =====\n");          summary(za_sel_d_1)
```

```
##
## ===== ZA Selic 1ª dif. (lag 1) =====
```

```
##
## ################################
```

```
## # Zivot-Andrews Unit Root Test #
## ################################
##
##
## Call:
## lm(formula = testmat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.09656 -0.21644 -0.00607  0.22356  0.89495
##
## Coefficients:
##             Estimate Std. Error t value             Pr(>|t|)
## (Intercept)  0.55417    0.50391   1.100              0.27243
## y.l1         0.62831    0.05444  11.541 < 0.0000000000000002 ***
## trend       -0.25346    0.08903  -2.847              0.00476 **
## y.dl1       -0.39041    0.05330  -7.325     0.00000000000282 ***
## du           1.44645    0.27631   5.235     0.00000033462981 ***
## dt           0.25367    0.08905   2.849              0.00473 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3573 on 267 degrees of freedom
##   (2 observations deleted due to missingness)
## Multiple R-squared:  0.4759, Adjusted R-squared:  0.4661
## F-statistic: 48.48 on 5 and 267 DF,  p-value: < 0.00000000000000022
##
##
## Teststatistic: -6.8273
## Critical values: 0.01= -5.57 0.05= -5.08 0.1= -4.82
##
## Potential break point at position: 8
```

5. Engle-Granger

```
## --- etapa 5: Engle-Granger - 3 regressões OLS + ADF nos resíduos ---
Y <- data.frame(
  log_ipca_indice = s_ipca,
  log_cambio      = s_cambio,
  selic           = s_selic
)

eg_ipca   <- lm(log_ipca_indice ~ log_cambio + selic, data = Y)
eg_cambio <- lm(log_cambio ~ log_ipca_indice + selic, data = Y)
eg_selic  <- lm(selic ~ log_ipca_indice + log_cambio, data = Y)

# ADF nos resíduos (none). CVs corretos = MacKinnon (1996), k = 3, com constante.
adf_eg_ipca <- ur.df(residuals(eg_ipca),   type = "none", selectlags = "AIC")
adf_eg_cam  <- ur.df(residuals(eg_cambio), type = "none", selectlags = "AIC")
adf_eg_sel  <- ur.df(residuals(eg_selic),  type = "none", selectlags = "AIC")

cv_eg <- c("1pct" = -4.29, "5pct" = -3.74, "10pct" = -3.45)
cat("Valores críticos de MacKinnon (1996) - Engle-Granger, k = 3, com constante:\n")
```

```
## Valores críticos de MacKinnon (1996) - Engle-Granger, k = 3, com constante:
```

```r
print(cv_eg)
```

```
##  1pct  5pct 10pct
## -4.29 -3.74 -3.45
```

5.1 Regressões OLS de cointegração — summary cru

```r
cat("===== EG: log(IPCA) ~ log(Câmbio) + Selic =====\n"); summary(eg_ipca)
```

```
## ===== EG: log(IPCA) ~ log(Câmbio) + Selic =====

##
## Call:
## lm(formula = log_ipca_indice ~ log_cambio + selic, data = Y)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.44839 -0.10737  0.02722  0.14369  0.33126
##
## Coefficients:
##              Estimate Std. Error t value            Pr(>|t|)
## (Intercept)  7.885361   0.045475  173.40 <0.0000000000000002 ***
## log_cambio   0.665192   0.027178   24.48 <0.0000000000000002 ***
## selic       -0.028313   0.002413  -11.73 <0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.184 on 273 degrees of freedom
## Multiple R-squared:  0.7558, Adjusted R-squared:  0.754
## F-statistic: 422.4 on 2 and 273 DF,  p-value: < 0.00000000000000022
```

```r
cat("\n===== EG: log(Câmbio) ~ log(IPCA) + Selic =====\n"); summary(eg_cambio)
```

```
##
## ===== EG: log(Câmbio) ~ log(IPCA) + Selic =====

##
## Call:
## lm(formula = log_cambio ~ log_ipca_indice + selic, data = Y)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.49160 -0.16533 -0.02697  0.09702  0.54518
##
## Coefficients:
##                  Estimate Std. Error t value             Pr(>|t|)
## (Intercept)     -7.747737   0.370033 -20.938 < 0.0000000000000002 ***
## log_ipca_indice  1.032697   0.042193  24.475 < 0.0000000000000002 ***
## selic            0.024975   0.003363   7.426     0.00000000000143 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2292 on 273 degrees of freedom
## Multiple R-squared:  0.6943, Adjusted R-squared:  0.6921
```

```
## F-statistic:   310 on 2 and 273 DF,  p-value: < 0.00000000000000022
```

```r
cat("\n===== EG: Selic ~ log(IPCA) + log(Câmbio) =====\n"); summary(eg_selic)
```

```
##
## ===== EG: Selic ~ log(IPCA) + log(Câmbio) =====
##
## Call:
## lm(formula = selic ~ log_ipca_indice + log_cambio, data = Y)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.1352  -2.3500   0.2556   2.6453   7.8621
##
## Coefficients:
##                 Estimate Std. Error t value             Pr(>|t|)
## (Intercept)     102.3721     7.5988  13.472 < 0.0000000000000002 ***
## log_ipca_indice -11.8426     1.0092 -11.735 < 0.0000000000000002 ***
## log_cambio        6.7289     0.9061   7.426     0.00000000000143 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.762 on 273 degrees of freedom
## Multiple R-squared:  0.3509, Adjusted R-squared:  0.3462
## F-statistic: 73.81 on 2 and 273 DF,  p-value: < 0.00000000000000022
```

5.2 ADF nos resíduos — summary cru (ler contra MacKinnon)

```r
cat("===== ADF resíduos - dep. log(IPCA) =====\n")
```

```
## ===== ADF resíduos - dep. log(IPCA) =====
```

```r
summary(adf_eg_ipca)
```

```
##
## ###############################################
## # Augmented Dickey-Fuller Test Unit Root Test #
## ###############################################
##
## Test regression none
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.098722 -0.010978  0.002015  0.018200  0.085525
##
## Coefficients:
##             Estimate Std. Error t value          Pr(>|t|)
## z.lag.1    -0.017592   0.008612  -2.043             0.042 *
## z.diff.lag  0.424239   0.054869   7.732 0.000000000000206 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 0.02579 on 272 degrees of freedom
## Multiple R-squared:  0.1854, Adjusted R-squared:  0.1794
## F-statistic: 30.96 on 2 and 272 DF,  p-value: 0.0000000000007712
##
##
## Value of test-statistic is: -2.0427
##
## Critical values for test statistics:
##       1pct  5pct 10pct
## tau1 -2.58 -1.95 -1.62
```

```r
cat("estatística tau1:", round(adf_eg_ipca@teststat["statistic","tau1"], 4),
    "| CV 5% MacKinnon:", cv_eg["5pct"], "\n")
```

```
## estatística tau1: -2.0427 | CV 5% MacKinnon: -3.74
```

```r
cat("\n===== ADF resíduos - dep. log(Câmbio) =====\n")
```

```
##
## ===== ADF resíduos - dep. log(Câmbio) =====
```

```r
summary(adf_eg_cam)
```

```
##
## ###############################################
## # Augmented Dickey-Fuller Test Unit Root Test #
## ###############################################
##
## Test regression none
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.106824 -0.025270 -0.003154  0.015630  0.144776
##
## Coefficients:
##             Estimate Std. Error t value        Pr(>|t|)
## z.lag.1    -0.024217   0.009467  -2.558          0.0111 *
## z.diff.lag  0.379435   0.055684   6.814 0.0000000000609 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.03553 on 272 degrees of freedom
## Multiple R-squared:  0.1591, Adjusted R-squared:  0.1529
## F-statistic: 25.73 on 2 and 272 DF,  p-value: 0.00000000005844
##
##
## Value of test-statistic is: -2.558
##
## Critical values for test statistics:
##       1pct  5pct 10pct
## tau1 -2.58 -1.95 -1.62
```

```r
cat("estatística tau1:", round(adf_eg_cam@teststat["statistic","tau1"], 4),
    "| CV 5% MacKinnon:", cv_eg["5pct"], "\n")
```

```
## estatística tau1: -2.558 | CV 5% MacKinnon: -3.74
```

```r
cat("\n===== ADF resíduos - dep. Selic =====\n")
```

```
##
## ===== ADF resíduos - dep. Selic =====
```

```r
summary(adf_eg_sel)
```

```
##
## ###############################################
## # Augmented Dickey-Fuller Test Unit Root Test #
## ###############################################
##
## Test regression none
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.8560 -0.2272  0.0304  0.2650  1.9108
##
## Coefficients:
##             Estimate Std. Error t value            Pr(>|t|)
## z.lag.1    -0.017421   0.007654  -2.276              0.0236 *
## z.diff.lag  0.530698   0.051185  10.368 <0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4693 on 272 degrees of freedom
## Multiple R-squared:  0.2866, Adjusted R-squared:  0.2814
## F-statistic: 54.65 on 2 and 272 DF,  p-value: < 0.00000000000000022
##
##
## Value of test-statistic is: -2.2761
##
## Critical values for test statistics:
##       1pct  5pct 10pct
## tau1 -2.58 -1.95 -1.62
```

```r
cat("estatística tau1:", round(adf_eg_sel@teststat["statistic","tau1"], 4),
    "| CV 5% MacKinnon:", cv_eg["5pct"], "\n")
```

```
## estatística tau1: -2.2761 | CV 5% MacKinnon: -3.74
```

6. Johansen

```r
## --- etapa 6: Johansen - seleção de defasagem e ca.jo (traço e máx autovalor) ---
var_sel <- VARselect(Y, lag.max = 12, type = "const")
```

```r
K_opt   <- max(as.integer(var_sel$selection["AIC(n)"]), 2)
cat("Defasagem selecionada (AIC) para Johansen: K =", K_opt, "\n\n")
```

```
## Defasagem selecionada (AIC) para Johansen: K = 5
```

```r
cat("Critérios de informação (VARselect):\n")
```

```
## Critérios de informação (VARselect):
```

```r
print(var_sel$selection)
```

```
## AIC(n)  HQ(n)  SC(n) FPE(n)
##      5      4      4      5
```

```r
jo_trace <- ca.jo(Y, type = "trace", K = K_opt,
                  ecdet = "const", spec = "longrun")
jo_eigen <- ca.jo(Y, type = "eigen", K = K_opt,
                  ecdet = "const", spec = "longrun")

rank_trace_5 <- sum(jo_trace@teststat > jo_trace@cval[, "5pct"])
rank_eigen_5 <- sum(jo_eigen@teststat > jo_eigen@cval[, "5pct"])
cat("\nRank pelo traço a 5%:", rank_trace_5,
    "| Rank pelo máximo autovalor a 5%:", rank_eigen_5, "\n")
```

```
##
## Rank pelo traço a 5%: 2 | Rank pelo máximo autovalor a 5%: 2
```

6.1 Johansen por traço — summary cru (autovetores e loadings)

```r
summary(jo_trace)
```

```
##
## ######################
## # Johansen-Procedure #
## ######################
##
## Test type: trace statistic , without linear trend and constant in cointegration
##
## Eigenvalues (lambda):
## [1] 0.17825152345423919686418 0.06116866462710524726676
## [3] 0.02031671731831163202342 0.00000000000000001061017
##
## Values of teststatistic and critical values of test:
##
##           test 10pct  5pct  1pct
## r <= 2 |  5.56  7.52  9.24 12.97
## r <= 1 | 22.67 17.85 19.96 24.60
## r = 0  | 75.87 32.00 34.91 41.07
##
## Eigenvectors, normalised to first column:
## (These are the cointegration relations)
##
##                     log_ipca_indice.l5 log_cambio.l5    selic.l5    constant
## log_ipca_indice.l5          1.00000000     1.0000000  1.00000000  1.00000000
## log_cambio.l5              -0.50313682    -0.3766748 -0.76616601  0.97110859
## selic.l5                   -0.03251781     0.4222753  0.02255715  0.04079072
```

```
## constant                 -9.52373960   -13.3112620 -7.71344432 -9.73089269
##
## Weights W:
## (This is the loading matrix)
##
##                    log_ipca_indice.l5 log_cambio.l5     selic.l5
## log_ipca_indice.d      -0.0012309951 -0.0001609359 -0.000664139
## log_cambio.d           -0.0006973477 -0.0018796890  0.024889249
## selic.d                 0.0729031082 -0.0442949400 -0.089948017
##                                    constant
## log_ipca_indice.d  0.000000000000000162287
## log_cambio.d       0.000000000000001479049
## selic.d           -0.000000000000005243935
```

6.2 Johansen por máximo autovalor — summary cru

```r
summary(jo_eigen)
```

```
##
## ######################
## # Johansen-Procedure #
## ######################
##
## Test type: maximal eigenvalue statistic (lambda max) , without linear trend and constant in cointegra
##
## Eigenvalues (lambda):
## [1] 0.17825152345423919686418 0.06116866462710524726676
## [3] 0.02031671731831163202342 0.00000000000000001061017
##
## Values of teststatistic and critical values of test:
##
##           test 10pct  5pct  1pct
## r <= 2 |  5.56  7.52  9.24 12.97
## r <= 1 | 17.11 13.75 15.67 20.20
## r = 0  | 53.20 19.77 22.00 26.81
##
## Eigenvectors, normalised to first column:
## (These are the cointegration relations)
##
##                     log_ipca_indice.l5 log_cambio.l5    selic.l5    constant
## log_ipca_indice.l5          1.00000000     1.0000000  1.00000000  1.00000000
## log_cambio.l5              -0.50313682    -0.3766748 -0.76616601  0.97110859
## selic.l5                   -0.03251781     0.4222753  0.02255715  0.04079072
## constant                   -9.52373960   -13.3112620 -7.71344432 -9.73089269
##
## Weights W:
## (This is the loading matrix)
##
##                    log_ipca_indice.l5 log_cambio.l5     selic.l5
## log_ipca_indice.d      -0.0012309951 -0.0001609359 -0.000664139
## log_cambio.d           -0.0006973477 -0.0018796890  0.024889249
## selic.d                 0.0729031082 -0.0442949400 -0.089948017
##                                    constant
## log_ipca_indice.d  0.000000000000000162287
```

```
## log_cambio.d       0.000000000000001479049
## selic.d           -0.000000000000005243935
```

7. VAR em 1ª diferença

```r
## --- etapa 7: VAR em 1ª diferença - seleção de p e estimação ---
dY <- data.frame(
  log_cambio      = diff(s_cambio),
  log_ipca_indice = diff(s_ipca),
  selic           = diff(s_selic)
)
dY_econ <- dY[, ORDEM_ECON]   # câmbio -> IPCA -> Selic

sel_var <- VARselect(dY_econ, lag.max = 12, type = "const")
p_var   <- max(as.integer(sel_var$selection["AIC(n)"]), 1)

cat("Ordem das variáveis no VAR:",
    paste(colnames(dY_econ), collapse = " -> "), "\n")
```

```
## Ordem das variáveis no VAR: log_cambio -> log_ipca_indice -> selic
```

```r
cat("Defasagem selecionada pelo AIC: p =", p_var, "\n\n")
```

```
## Defasagem selecionada pelo AIC: p = 4
```

```r
cat("Critérios de informação (VARselect):\n")
```

```
## Critérios de informação (VARselect):
```

```r
print(sel_var$selection)
```

```
## AIC(n)  HQ(n)  SC(n) FPE(n)
##      4      3      3      4
```

```r
var_fit <- VAR(dY_econ, p = p_var, type = "const")
```

7.1 summary(VAR) — saída crua

```r
summary(var_fit)
```

```
##
## VAR Estimation Results:
## =========================
## Endogenous variables: log_cambio, log_ipca_indice, selic
## Deterministic variables: const
## Sample size: 271
## Log Likelihood: 1674.421
## Roots of the characteristic polynomial:
## 0.8202 0.7444 0.7444 0.6874 0.6874 0.5953 0.5953 0.554 0.554 0.5126 0.5126 0.4513
## Call:
## VAR(y = dY_econ, p = p_var, type = "const")
##
##
## Estimation results for equation log_cambio:
## ===========================================
```

```
## log_cambio = log_cambio.l1 + log_ipca_indice.l1 + selic.l1 + log_cambio.l2 + log_ipca_indice.l2 + sel
##
##                     Estimate Std. Error t value    Pr(>|t|)
## log_cambio.l1       0.3549433  0.0624154   5.687 0.000000035 ***
## log_ipca_indice.l1  0.1101646  0.7707920   0.143       0.886
## selic.l1            0.0030420  0.0058931   0.516       0.606
## log_cambio.l2      -0.0927327  0.0662833  -1.399       0.163
## log_ipca_indice.l2 -0.5869891  0.8633510  -0.680       0.497
## selic.l2           -0.0018104  0.0057772  -0.313       0.754
## log_cambio.l3       0.0441540  0.0669723   0.659       0.510
## log_ipca_indice.l3  0.8514769  0.8659203   0.983       0.326
## selic.l3            0.0053416  0.0056692   0.942       0.347
## log_cambio.l4      -0.0482857  0.0623085  -0.775       0.439
## log_ipca_indice.l4 -1.0862145  0.7709851  -1.409       0.160
## selic.l4           -0.0006228  0.0056040  -0.111       0.912
## const               0.0052755  0.0050486   1.045       0.297
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.0338 on 258 degrees of freedom
## Multiple R-Squared: 0.1262,  Adjusted R-squared: 0.08559
## F-statistic: 3.106 on 12 and 258 DF,  p-value: 0.0003929
##
##
## Estimation results for equation log_ipca_indice:
## ================================================
## log_ipca_indice = log_cambio.l1 + log_ipca_indice.l1 + selic.l1 + log_cambio.l2 + log_ipca_indice.l2
##
##                      Estimate  Std. Error t value          Pr(>|t|)
## log_cambio.l1       0.00204292  0.00500775   0.408            0.6836
## log_ipca_indice.l1  0.48623754  0.06184270   7.862 0.000000000000103 ***
## selic.l1            0.00068490  0.00047282   1.449            0.1487
## log_cambio.l2       0.00147842  0.00531808   0.278            0.7812
## log_ipca_indice.l2  0.05069004  0.06926896   0.732            0.4650
## selic.l2            0.00048952  0.00046352   1.056            0.2919
## log_cambio.l3      -0.00073324  0.00537337  -0.136            0.8916
## log_ipca_indice.l3 -0.02461471  0.06947510  -0.354            0.7234
## selic.l3            0.00008533  0.00045485   0.188            0.8513
## log_cambio.l4       0.01248351  0.00499918   2.497            0.0131 *
## log_ipca_indice.l4 -0.07670887  0.06185820  -1.240            0.2161
## selic.l4           -0.00043537  0.00044963  -0.968            0.3338
## const               0.00256194  0.00040506   6.325 0.000000001111975 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.002712 on 258 degrees of freedom
## Multiple R-Squared: 0.3323,  Adjusted R-squared: 0.3012
## F-statistic:  10.7 on 12 and 258 DF,  p-value: < 0.00000000000000022
##
##
## Estimation results for equation selic:
## ======================================
```

```
## selic = log_cambio.l1 + log_ipca_indice.l1 + selic.l1 + log_cambio.l2 + log_ipca_indice.l2 + selic.l2
##
##                      Estimate Std. Error t value    Pr(>|t|)
## log_cambio.l1       -1.12425    0.64377  -1.746     0.08194 .
## log_ipca_indice.l1  17.13762    7.95016   2.156     0.03204 *
## selic.l1             0.26925    0.06078   4.430 0.000013955 ***
## log_cambio.l2        1.83621    0.68366   2.686     0.00770 **
## log_ipca_indice.l2   5.79937    8.90484   0.651     0.51546
## selic.l2             0.32561    0.05959   5.464 0.000000109 ***
## log_cambio.l3       -0.88808    0.69077  -1.286     0.19973
## log_ipca_indice.l3  11.30763    8.93134   1.266     0.20663
## selic.l3             0.30656    0.05847   5.243 0.000000330 ***
## log_cambio.l4        0.12276    0.64267   0.191     0.84866
## log_ipca_indice.l4 -12.85473    7.95216  -1.617     0.10721
## selic.l4            -0.18488    0.05780  -3.199     0.00155 **
## const               -0.10735    0.05207  -2.062     0.04025 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.3486 on 258 degrees of freedom
## Multiple R-Squared: 0.5177,  Adjusted R-squared: 0.4953
## F-statistic: 23.08 on 12 and 258 DF,  p-value: < 0.00000000000000022
##
##
##
## Covariance matrix of residuals:
##                  log_cambio log_ipca_indice      selic
## log_cambio       0.00114238    -0.000011477 0.00001277
## log_ipca_indice -0.00001148     0.000007354 0.00006731
## selic            0.00001277     0.000067308 0.12153138
##
## Correlation matrix of residuals:
##                 log_cambio log_ipca_indice    selic
## log_cambio        1.000000         -0.1252 0.001083
## log_ipca_indice  -0.125223          1.0000 0.071198
## selic             0.001083          0.0712 1.000000
```

7.2 Matrizes de covariância e correlação dos resíduos

```r
sv <- summary(var_fit)
cat("Matriz de covariância dos resíduos:\n")
```

```
## Matriz de covariância dos resíduos:
```

```r
print(round(sv$covres, 6))
```

```
##                 log_cambio log_ipca_indice    selic
## log_cambio        0.001142       -0.000011 0.000013
## log_ipca_indice  -0.000011        0.000007 0.000067
## selic             0.000013        0.000067 0.121531
```

```r
cat("\nMatriz de correlação dos resíduos:\n")
```

```
##
```

```
## Matriz de correlação dos resíduos:
```

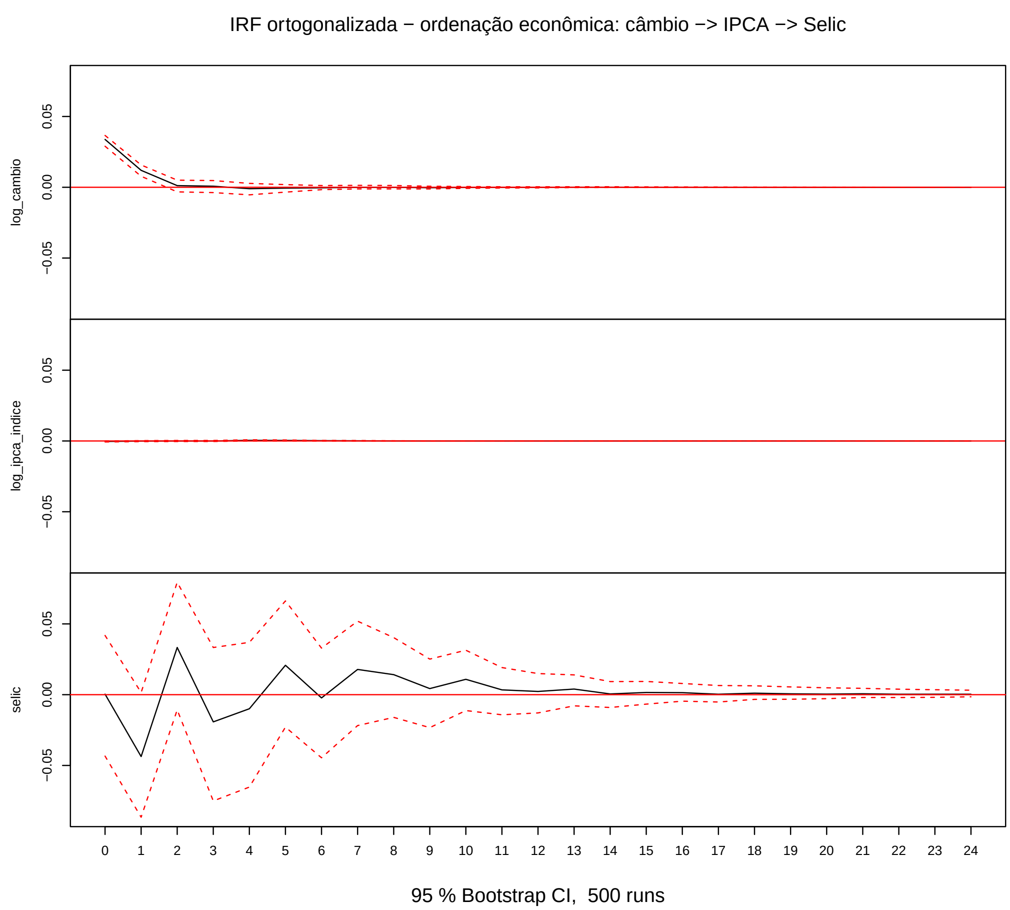
```r
print(round(sv$corres, 4))
```

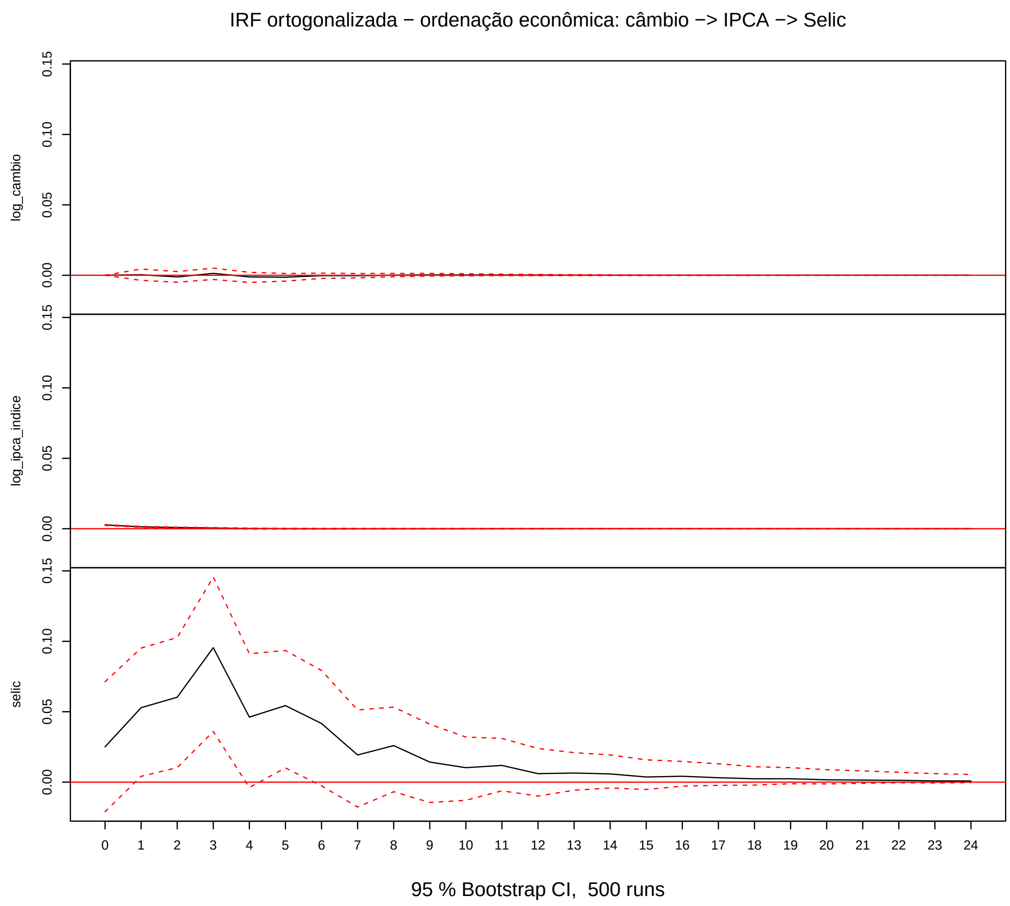
```
##                 log_cambio log_ipca_indice  selic
## log_cambio          1.0000         -0.1252 0.0011
## log_ipca_indice    -0.1252          1.0000 0.0712
## selic               0.0011          0.0712 1.0000
```

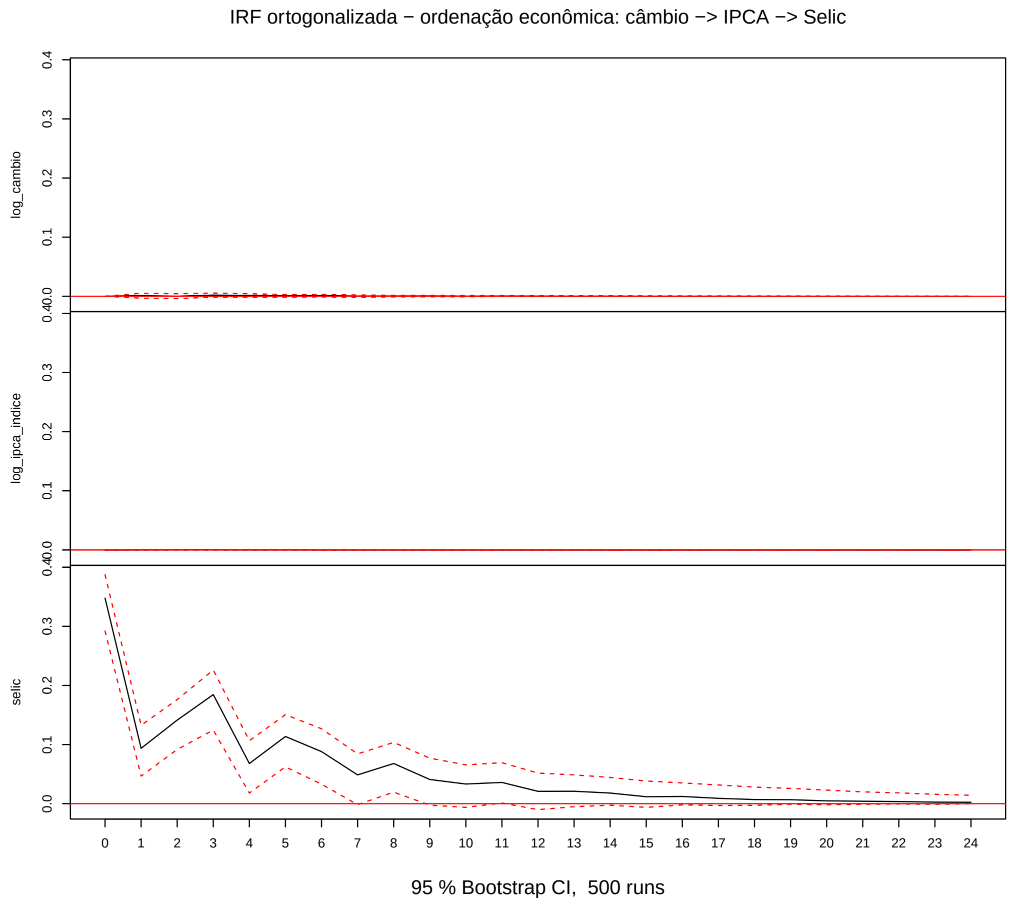
8. IRFs ortogonalizadas, FEVD e IRF acumulada do VECM

8.1 IRF — ordenação econômica (câmbio → IPCA → Selic)

```r
## --- etapa 8: IRF VAR ortogonalizada, ordenação econômica ---
set.seed(SEED)
irf_econ <- irf(var_fit, n.ahead = HORIZONTE_IR,
                ortho = TRUE, boot = TRUE, runs = N_BOOT, ci = 0.95)
plot(irf_econ,
     main = "IRF ortogonalizada - ordenação econômica: câmbio -> IPCA -> Selic")
```

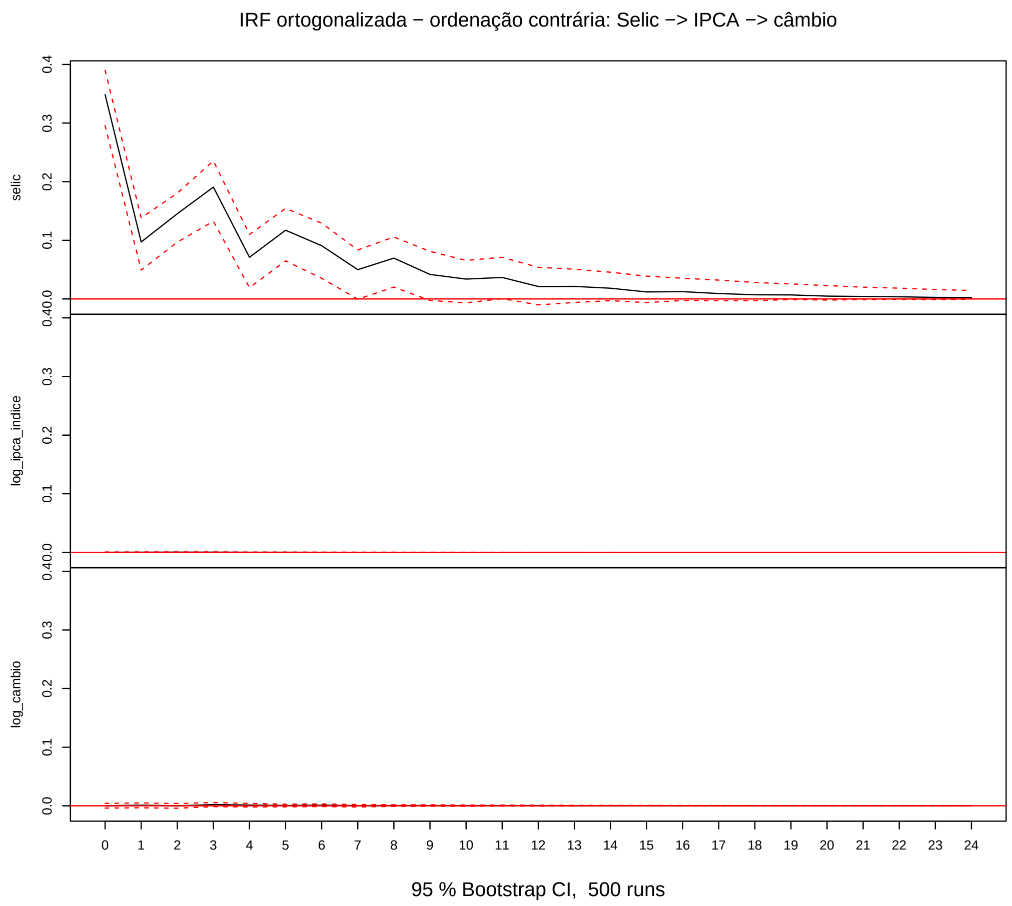



IRF ortogonalizada − ordenação econômica: câmbio −> IPCA −> Selic
log_cambio
log_ipca_indice
selic
0.00
0.05
0.10
0.15
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24
95 % Bootstrap CI, 500 runs

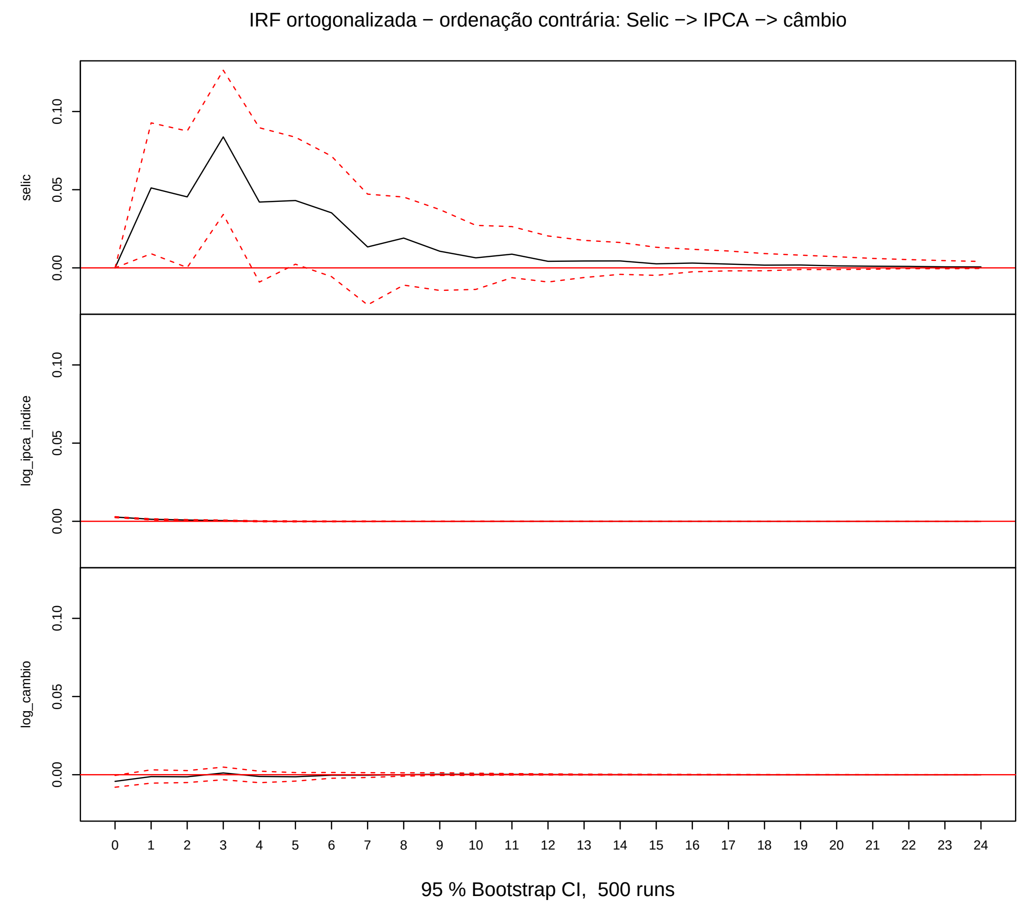


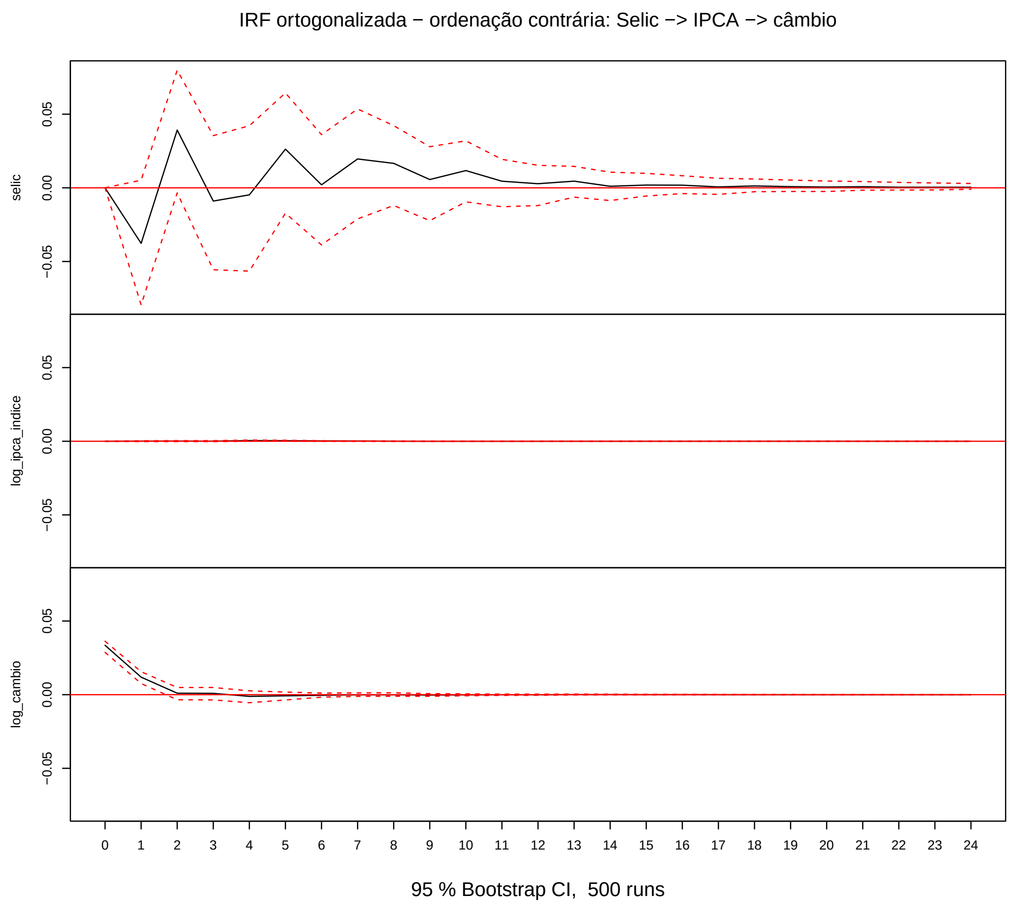
8.2 IRF — ordenação contrária (Selic → IPCA → câmbio)

```
## --- etapa 8: IRF VAR ortogonalizada, ordenação contrária (robustez) ---
dY_cont  <- dY[, ORDEM_CONT]
var_cont <- VAR(dY_cont, p = p_var, type = "const")

set.seed(SEED)
irf_cont <- irf(var_cont, n.ahead = HORIZONTE_IR,
                ortho = TRUE, boot = TRUE, runs = N_BOOT, ci = 0.95)
plot(irf_cont,
     main = "IRF ortogonalizada - ordenação contrária: Selic -> IPCA -> câmbio")
```



IRF ortogonalizada − ordenação contrária: Selic −> IPCA −> câmbio
selic
log_ipca_indice
log_cambio
0.0
0.1
0.2
0.3
0.4
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24
95 % Bootstrap CI, 500 runs


IRF ortogonalizada − ordenação contrária: Selic −> IPCA −> câmbio
selic
log_ipca_indice
log_cambio
0.00
0.05
0.10
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24
95 % Bootstrap CI, 500 runs



8.3 FEVD do IPCA

```
## --- etapa 8: FEVD do VAR (em diferença) ---
fevd_var <- fevd(var_fit, n.ahead = HORIZONTE_IR)
cat("===== FEVD (VAR) - fração da variância de previsão do IPCA por choque =====\n")
```

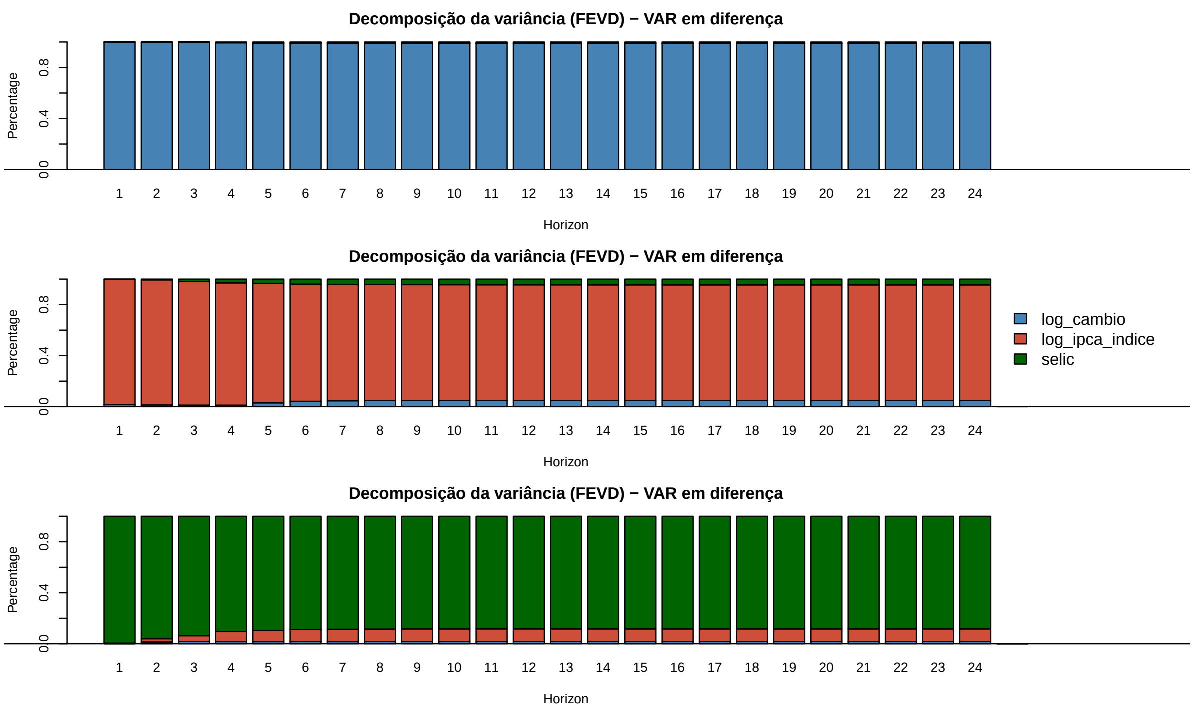
```
## ===== FEVD (VAR) - fração da variância de previsão do IPCA por choque =====
```

```
print(round(fevd_var$log_ipca_indice, 4))
```

```
##       log_cambio log_ipca_indice  selic
##  [1,]     0.0157          0.9843 0.0000
##  [2,]     0.0136          0.9803 0.0062
##  [3,]     0.0125          0.9694 0.0181
##  [4,]     0.0121          0.9581 0.0298
##  [5,]     0.0299          0.9355 0.0346
##  [6,]     0.0423          0.9186 0.0391
##  [7,]     0.0461          0.9128 0.0411
##  [8,]     0.0477          0.9099 0.0424
##  [9,]     0.0477          0.9086 0.0436
## [10,]     0.0477          0.9081 0.0442
## [11,]     0.0477          0.9077 0.0447
## [12,]     0.0477          0.9074 0.0449
## [13,]     0.0477          0.9072 0.0451
```

```
## [14,]      0.0477          0.9071 0.0452
## [15,]      0.0477          0.9070 0.0453
## [16,]      0.0477          0.9070 0.0453
## [17,]      0.0477          0.9069 0.0454
## [18,]      0.0477          0.9069 0.0454
## [19,]      0.0477          0.9069 0.0454
## [20,]      0.0477          0.9069 0.0454
## [21,]      0.0477          0.9069 0.0454
## [22,]      0.0477          0.9069 0.0455
## [23,]      0.0477          0.9069 0.0455
## [24,]      0.0477          0.9069 0.0455
```

```
op <- par(no.readonly = TRUE)
par(mar = c(4, 4, 3, 8), xpd = TRUE)
plot(fevd_var, main = "Decomposição da variância (FEVD) - VAR em diferença",
     col = c("steelblue", "tomato3", "darkgreen"), legend = FALSE)
legend("right", inset = c(-0.18, 0), legend = colnames(fevd_var[[1]]),
       fill = c("steelblue", "tomato3", "darkgreen"), bty = "n", cex = 0.85, xpd = TRUE)
```




```
par(op)
```

8.4 IRF acumulada do VECM via vec2var (choque no câmbio)

```
## --- etapa 8: ca.jo -> vec2var -> irf acumulada (ordenação econômica) ---
rank_coint <- max(sum(jo_trace@teststat > jo_trace@cval[, "5pct"]), 1)

Y_econ_nivel  <- Y[, ORDEM_ECON]
jo_trace_econ <- ca.jo(Y_econ_nivel, type = "trace", K = K_opt,
                       ecdet = "const", spec = "longrun")
vecm_var_econ <- vec2var(jo_trace_econ, r = rank_coint)
```

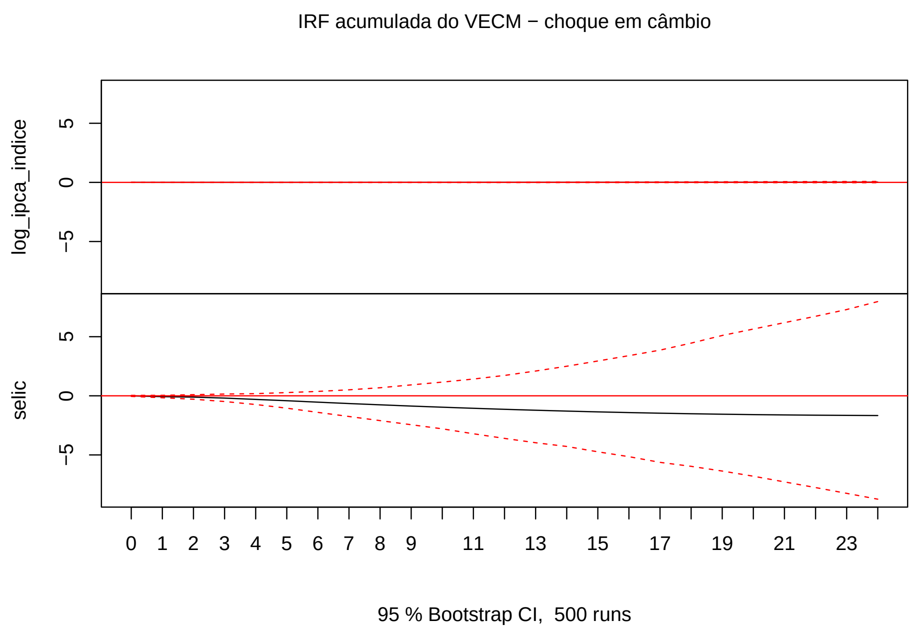
```r
set.seed(SEED)
irf_vecm <- irf(vecm_var_econ, impulse = "log_cambio",
                response = c("log_ipca_indice", "selic"),
                n.ahead = HORIZONTE_IR, ortho = TRUE, boot = TRUE,
                runs = N_BOOT, ci = 0.95, cumulative = TRUE)

cat("===== IRF acumulada do VECM - choque em log_cambio (estimativa pontual) =====\n")
```

```
## ===== IRF acumulada do VECM - choque em log_cambio (estimativa pontual) =====
```

```r
print(round(irf_vecm$irf$log_cambio, 6))
```

```
##       log_ipca_indice     selic
##  [1,]       -0.000307 -0.007889
##  [2,]       -0.000676 -0.069029
##  [3,]       -0.001029 -0.111741
##  [4,]       -0.001360 -0.192219
##  [5,]       -0.001235 -0.305309
##  [6,]       -0.000736 -0.414257
##  [7,]       -0.000032 -0.537762
##  [8,]        0.000804 -0.654464
##  [9,]        0.001683 -0.761859
## [10,]        0.002577 -0.867656
## [11,]        0.003498 -0.964029
## [12,]        0.004458 -1.055129
## [13,]        0.005473 -1.141406
## [14,]        0.006551 -1.220005
## [15,]        0.007697 -1.292936
## [16,]        0.008912 -1.359146
## [17,]        0.010196 -1.418193
## [18,]        0.011549 -1.470934
## [19,]        0.012970 -1.516882
## [20,]        0.014458 -1.556394
## [21,]        0.016012 -1.589892
## [22,]        0.017630 -1.617424
## [23,]        0.019311 -1.639483
## [24,]        0.021055 -1.656403
## [25,]        0.022859 -1.668492
```

```r
plot(irf_vecm, main = "IRF acumulada do VECM - choque em câmbio")
```



8.5 FEVD do IPCA pelo VECM e IRF acumulada sob ordenação invertida

```r
## --- etapa 8: FEVD do IPCA via VECM + robustez da ordenação ---
fevd_obj <- fevd(vecm_var_econ, n.ahead = HORIZONTE_IR)
cat("===== FEVD (VECM) - fração da variância de previsão do IPCA por choque =====\n")
```

```
## ===== FEVD (VECM) - fração da variância de previsão do IPCA por choque =====
```

```r
print(round(fevd_obj$log_ipca_indice, 4))
```

```
##       log_cambio log_ipca_indice  selic
##  [1,]     0.0137          0.9863 0.0000
##  [2,]     0.0105          0.9870 0.0025
##  [3,]     0.0081          0.9833 0.0086
##  [4,]     0.0067          0.9764 0.0169
##  [5,]     0.0050          0.9700 0.0251
##  [6,]     0.0059          0.9611 0.0329
##  [7,]     0.0082          0.9524 0.0394
##  [8,]     0.0109          0.9447 0.0444
##  [9,]     0.0133          0.9385 0.0482
## [10,]     0.0153          0.9341 0.0507
## [11,]     0.0171          0.9308 0.0521
## [12,]     0.0188          0.9285 0.0527
## [13,]     0.0207          0.9269 0.0524
## [14,]     0.0228          0.9257 0.0515
## [15,]     0.0250          0.9248 0.0502
## [16,]     0.0274          0.9241 0.0485
## [17,]     0.0301          0.9233 0.0466
## [18,]     0.0330          0.9224 0.0447
## [19,]     0.0360          0.9212 0.0428
## [20,]     0.0393          0.9198 0.0409
```

```
## [21,]     0.0427          0.9181 0.0392
## [22,]     0.0463          0.9160 0.0377
## [23,]     0.0501          0.9134 0.0365
## [24,]     0.0540          0.9105 0.0355
```

```r
# Ordenação invertida: Selic -> IPCA -> câmbio (câmbio mais endógeno)
jo_trace_rev <- ca.jo(Y[, ORDEM_CONT], type = "trace", K = K_opt,
                      ecdet = "const", spec = "longrun")
vecm_var_rev <- vec2var(jo_trace_rev, r = rank_coint)

set.seed(SEED)
irf_cu_rev <- irf(vecm_var_rev, impulse = "log_cambio", response = "log_ipca_indice",
                  n.ahead = HORIZONTE_IR, ortho = TRUE, boot = TRUE,
                  runs = N_BOOT, ci = 0.95, cumulative = TRUE)
cat("\nPass-through acumulado do IPCA em h=24 - baseline:",
    round(tail(irf_vecm$irf$log_cambio[, "log_ipca_indice"], 1), 4),
    "| invertida:",
    round(tail(irf_cu_rev$irf$log_cambio[, "log_ipca_indice"], 1), 4), "\n")
```

```
##
## Pass-through acumulado do IPCA em h=24 - baseline: 0.0229 | invertida: 0.036
```

9. Diagnósticos do VAR e do VECM

9.1 Diagnósticos do VAR

```r
## --- etapa 9: diagnósticos do VAR ---
pt_test   <- serial.test(var_fit, lags.pt = 12, type = "PT.adjusted")
norm_test <- normality.test(var_fit)
arch_var  <- arch.test(var_fit, lags.multi = 5, multivariate.only = TRUE)

resid_var  <- residuals(var_fit)
fitted_var <- fitted(var_fit)
bp_results <- lapply(seq_len(ncol(resid_var)), function(i) {
  lmtest::bptest(lm(resid_var[, i]^2 ~ fitted_var[, i] + I(fitted_var[, i]^2)))
})
names(bp_results) <- colnames(resid_var)

cat("===== Portmanteau ajustado (VAR), lags.pt = 12 =====\n"); print(pt_test$serial)
```

```
## ===== Portmanteau ajustado (VAR), lags.pt = 12 =====

##
##  Portmanteau Test (adjusted)
##
## data:  Residuals of VAR object var_fit
## Chi-squared = 88.437, df = 72, p-value = 0.09141
```

```r
cat("\n===== Jarque-Bera multivariado (VAR) =====\n");          print(norm_test$jb.mul)
```

```
##
## ===== Jarque-Bera multivariado (VAR) =====

## $JB
##
##  JB-Test (multivariate)
```

```
##
## data:  Residuals of VAR object var_fit
## Chi-squared = 228.85, df = 6, p-value < 0.00000000000000022
##
##
## $Skewness
##
##  Skewness only (multivariate)
##
## data:  Residuals of VAR object var_fit
## Chi-squared = 30.866, df = 3, p-value = 0.000000907
##
##
## $Kurtosis
##
##  Kurtosis only (multivariate)
##
## data:  Residuals of VAR object var_fit
## Chi-squared = 197.99, df = 3, p-value < 0.00000000000000022
```

```
cat("\n===== ARCH multivariado (VAR), lags.multi = 5 =====\n"); print(arch_var$arch.mul)
```

```
##
## ===== ARCH multivariado (VAR), lags.multi = 5 =====
```

```
##
##  ARCH (multivariate)
##
## data:  Residuals of VAR object var_fit
## Chi-squared = 324.94, df = 180, p-value = 0.0000000002085
```

```
cat("\n===== Breusch-Pagan por equação (VAR) =====\n")
```

```
##
## ===== Breusch-Pagan por equação (VAR) =====
```

```
for (nm in names(bp_results)) { cat("\n", nm, "\n", sep = ""); print(bp_results[[nm]]) }
```

```
##
## log_cambio
##
##  studentized Breusch-Pagan test
##
## data:  lm(resid_var[, i]^2 ~ fitted_var[, i] + I(fitted_var[, i]^2))
## BP = 55.019, df = 2, p-value = 0.000000000001129
##
##
## log_ipca_indice
##
##  studentized Breusch-Pagan test
##
## data:  lm(resid_var[, i]^2 ~ fitted_var[, i] + I(fitted_var[, i]^2))
## BP = 0.42529, df = 2, p-value = 0.8084
##
##
## selic
##
```

```
##  studentized Breusch-Pagan test
##
## data:  lm(resid_var[, i]^2 ~ fitted_var[, i] + I(fitted_var[, i]^2))
## BP = 5.6938, df = 2, p-value = 0.05803
```

```r
cat("\n===== Raízes do polinômio característico (VAR) =====\n")
```

```
##
## ===== Raízes do polinômio característico (VAR) =====
```

```r
print(round(roots(var_fit), 4))
```

```
##  [1] 0.8202 0.7444 0.7444 0.6874 0.6874 0.5953 0.5953 0.5540 0.5540 0.5126
## [11] 0.5126 0.4513
```

```r
cat("Sistema VAR estável (todas as raízes < 1):", all(roots(var_fit) < 1), "\n")
```

```
## Sistema VAR estável (todas as raízes < 1): TRUE
```

9.2 Diagnósticos do VECM

```r
## --- etapa 9: diagnósticos do VECM (base e robustez K=6) ---
vecm_as_var <- vec2var(jo_trace, r = rank_coint)

pt_vecm   <- serial.test(vecm_as_var, lags.pt = 12, type = "PT.adjusted")
norm_vecm <- normality.test(vecm_as_var)

resid_vecm  <- residuals(vecm_as_var)
fitted_vecm <- fitted(vecm_as_var)
bp_vecm <- lapply(seq_len(ncol(resid_vecm)), function(i) {
  lmtest::bptest(lm(resid_vecm[, i]^2 ~ fitted_vecm[, i] + I(fitted_vecm[, i]^2)))
})
names(bp_vecm) <- colnames(resid_vecm)

# Raízes via matriz companheira
A_mats <- vecm_as_var$A
Kv <- vecm_as_var$K; pv <- vecm_as_var$p
companion <- matrix(0, nrow = Kv * pv, ncol = Kv * pv)
for (i in seq_len(pv)) companion[1:Kv, ((i-1)*Kv + 1):(i*Kv)] <- A_mats[[i]]
if (pv > 1) companion[(Kv+1):(Kv*pv), 1:(Kv*(pv-1))] <- diag(Kv * (pv-1))
raizes_vecm <- Mod(eigen(companion, only.values = TRUE)$values)

# Robustez: VECM com K = 6
jo_trace_6   <- ca.jo(Y, type = "trace", K = 6, ecdet = "const", spec = "longrun")
vecm6_as_var <- vec2var(jo_trace_6, r = rank_coint)
pt_vecm6     <- serial.test(vecm6_as_var, lags.pt = 12, type = "PT.adjusted")

cat("===== Portmanteau ajustado (VECM-base), lags.pt = 12 =====\n"); print(pt_vecm$serial)
```

```
## ===== Portmanteau ajustado (VECM-base), lags.pt = 12 =====
```

```
##
##  Portmanteau Test (adjusted)
##
## data:  Residuals of VAR object vecm_as_var
## Chi-squared = 90.2, df = 66, p-value = 0.02562
```

```r
cat("\n===== Portmanteau ajustado (VECM K = 6), lags.pt = 12 =====\n"); print(pt_vecm6$serial)
```

```
##
## ===== Portmanteau ajustado (VECM K = 6), lags.pt = 12 =====
##
##  Portmanteau Test (adjusted)
##
## data:  Residuals of VAR object vecm6_as_var
## Chi-squared = 89.764, df = 57, p-value = 0.003652
```

```r
cat("\n===== Jarque-Bera multivariado (VECM) =====\n");                print(norm_vecm$jb.mul)
```

```
##
## ===== Jarque-Bera multivariado (VECM) =====
## $JB
##
##  JB-Test (multivariate)
##
## data:  Residuals of VAR object vecm_as_var
## Chi-squared = 176.69, df = 6, p-value < 0.00000000000000022
##
##
## $Skewness
##
##  Skewness only (multivariate)
##
## data:  Residuals of VAR object vecm_as_var
## Chi-squared = 22.359, df = 3, p-value = 0.00005492
##
##
## $Kurtosis
##
##  Kurtosis only (multivariate)
##
## data:  Residuals of VAR object vecm_as_var
## Chi-squared = 154.33, df = 3, p-value < 0.00000000000000022
```

```r
cat("\n===== Breusch-Pagan por equação (VECM) =====\n")
```

```
##
## ===== Breusch-Pagan por equação (VECM) =====
```

```r
for (nm in names(bp_vecm)) { cat("\n", nm, "\n", sep = ""); print(bp_vecm[[nm]]) }
```

```
##
## resids of log_ipca_indice
##
##  studentized Breusch-Pagan test
##
## data:  lm(resid_vecm[, i]^2 ~ fitted_vecm[, i] + I(fitted_vecm[, i]^2))
## BP = 8.3064, df = 2, p-value = 0.01571
##
##
## resids of log_cambio
##
```

```
##  studentized Breusch-Pagan test
##
## data:  lm(resid_vecm[, i]^2 ~ fitted_vecm[, i] + I(fitted_vecm[, i]^2))
## BP = 1.5087, df = 2, p-value = 0.4703
##
##
## resids of selic
##
##  studentized Breusch-Pagan test
##
## data:  lm(resid_vecm[, i]^2 ~ fitted_vecm[, i] + I(fitted_vecm[, i]^2))
## BP = 30.5, df = 2, p-value = 0.0000002383
```

```
cat("\n===== Raízes (VECM) =====\n")
```

```
##
## ===== Raízes (VECM) =====
```

```
print(round(sort(raizes_vecm, decreasing = TRUE), 4))
```

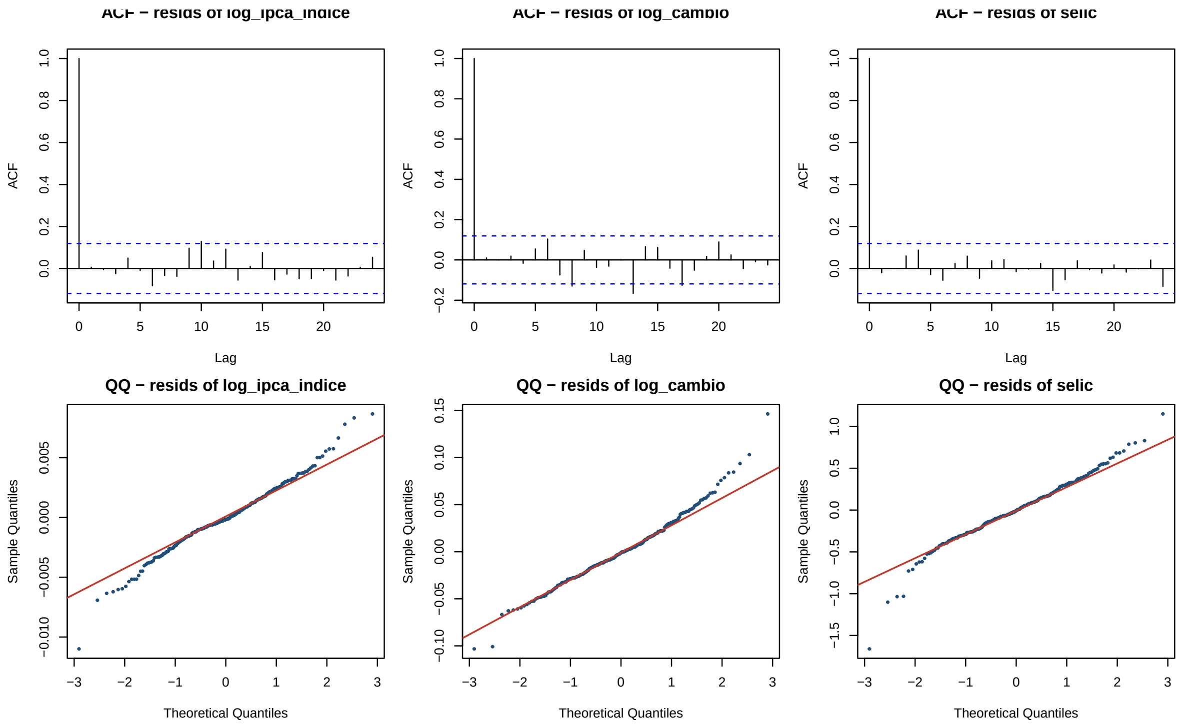
```
##  [1] 1.0000 0.9983 0.9171 0.9171 0.7307 0.7307 0.6611 0.6611 0.6034 0.6034
## [11] 0.5421 0.5421 0.5221 0.5221 0.3841
```

```
cat("max|raiz| =", round(max(raizes_vecm), 4),
    "| k - r =", ncol(Y) - rank_coint, "tendência(s) estocástica(s) comum(ns)\n")
```

```
## max|raiz| = 1 | k - r = 1 tendência(s) estocástica(s) comum(ns)
```

9.3 Diagnóstico visual — ACF e QQ dos resíduos do VECM

```
nm <- colnames(resid_vecm)
par(mfrow = c(2, 3), mar = c(4, 4, 2.5, 1))
for (j in seq_len(ncol(resid_vecm)))
  acf(resid_vecm[, j], lag.max = 24, main = paste("ACF -", nm[j]))
for (j in seq_len(ncol(resid_vecm))) {
  qqnorm(resid_vecm[, j], main = paste("QQ -", nm[j]), pch = 20, cex = 0.6, col = "#1f4e79")
  qqline(resid_vecm[, j], col = "#c0392b", lwd = 1.4)
}
```



9.4 VECM via cajorls — beta, alpha e summary cru

```r
## --- etapa 9: estimação do VECM por cajorls (rank do traço) ---
vecm_fit <- cajorls(jo_trace, r = rank_coint)

cat("Rank de cointegração adotado:", rank_coint, "\n")
```

```
## Rank de cointegração adotado: 2
```

```r
cat("\n===== Vetor(es) de cointegração - beta =====\n")
```

```
##
## ===== Vetor(es) de cointegração - beta =====
```

```r
print(round(vecm_fit$beta, 4))
```

```
##                        ect1     ect2
## log_ipca_indice.l5   1.0000   0.0000
## log_cambio.l5        0.0000   1.0000
## selic.l5             1.7769   3.5963
## constant           -24.5926 -29.9499
```

```r
alpha <- coef(vecm_fit$rlm)[1:rank_coint, , drop = FALSE]
cat("\n===== Coeficientes de ajustamento - alpha =====\n")
```

```
##
## ===== Coeficientes de ajustamento - alpha =====
```

```r
print(round(alpha, 4))
```

```
##      log_ipca_indice.d log_cambio.d selic.d
## ect1           -0.0014      -0.0026  0.0286
## ect2            0.0007       0.0011 -0.0200
```

```r
cat("\n===== summary completo das equações do VECM (rlm) =====\n")
```

```
##
## ===== summary completo das equações do VECM (rlm) =====
```

```r
summary(vecm_fit$rlm)
```

```
## Response log_ipca_indice.d :
##
## Call:
## lm(formula = log_ipca_indice.d ~ ect1 + ect2 + log_ipca_indice.dl1 +
##     log_cambio.dl1 + selic.dl1 + log_ipca_indice.dl2 + log_cambio.dl2 +
##     selic.dl2 + log_ipca_indice.dl3 + log_cambio.dl3 + selic.dl3 +
##     log_ipca_indice.dl4 + log_cambio.dl4 + selic.dl4 - 1, data = data.mat)
##
## Residuals:
##        Min         1Q     Median         3Q        Max
## -0.0109938 -0.0013766 -0.0001643  0.0015508  0.0086590
##
## Coefficients:
##                        Estimate  Std. Error t value          Pr(>|t|)
## ect1                -0.00139193  0.00021547  -6.460 0.000000000521214 ***
## ect2                 0.00067998  0.00010297   6.604 0.000000000228526 ***
## log_ipca_indice.dl1  0.47270843  0.06188049   7.639 0.000000000000433 ***
## log_cambio.dl1       0.00271717  0.00499638   0.544            0.5870
## selic.dl1            0.00071122  0.00049187   1.446            0.1494
## log_ipca_indice.dl2  0.04405893  0.06894533   0.639            0.5234
## log_cambio.dl2       0.00211203  0.00531320   0.398            0.6913
## selic.dl2            0.00053739  0.00046887   1.146            0.2528
## log_ipca_indice.dl3 -0.03026812  0.06915062  -0.438            0.6620
## log_cambio.dl3      -0.00005437  0.00534824  -0.010            0.9919
## selic.dl3            0.00016597  0.00045411   0.365            0.7151
## log_ipca_indice.dl4 -0.09164536  0.06245038  -1.467            0.1435
## log_cambio.dl4       0.01326454  0.00500143   2.652            0.0085 **
## selic.dl4           -0.00036561  0.00045060  -0.811            0.4179
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.002696 on 257 degrees of freedom
## Multiple R-squared:  0.7766, Adjusted R-squared:  0.7645
## F-statistic: 63.83 on 14 and 257 DF,  p-value: < 0.00000000000000022
##
##
## Response log_cambio.d :
##
## Call:
## lm(formula = log_cambio.d ~ ect1 + ect2 + log_ipca_indice.dl1 +
##     log_cambio.dl1 + selic.dl1 + log_ipca_indice.dl2 + log_cambio.dl2 +
##     selic.dl2 + log_ipca_indice.dl3 + log_cambio.dl3 + selic.dl3 +
##     log_ipca_indice.dl4 + log_cambio.dl4 + selic.dl4 - 1, data = data.mat)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.103236 -0.020492 -0.001125  0.018606  0.146363
```

```
##
## Coefficients:
##                       Estimate  Std. Error t value     Pr(>|t|)
## ect1               -0.00257704  0.00270040  -0.954        0.341
## ect2                0.00105889  0.00129043   0.821        0.413
## log_ipca_indice.dl1  0.18794021  0.77552418   0.242        0.809
## log_cambio.dl1       0.34927894  0.06261765   5.578 0.0000000616 ***
## selic.dl1            0.00009176  0.00616445   0.015        0.988
## log_ipca_indice.dl2 -0.52599415  0.86406501  -0.609        0.543
## log_cambio.dl2      -0.10032473  0.06658829  -1.507        0.133
## selic.dl2           -0.00355544  0.00587620  -0.605        0.546
## log_ipca_indice.dl3  0.94479963  0.86663777   1.090        0.277
## log_cambio.dl3       0.04064460  0.06702746   0.606        0.545
## selic.dl3            0.00472859  0.00569118   0.831        0.407
## log_ipca_indice.dl4 -0.83391300  0.78266633  -1.065        0.288
## log_cambio.dl4      -0.05646625  0.06268093  -0.901        0.369
## selic.dl4            0.00034158  0.00564725   0.060        0.952
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.03379 on 257 degrees of freedom
## Multiple R-squared:  0.1337, Adjusted R-squared:  0.08655
## F-statistic: 2.834 on 14 and 257 DF,  p-value: 0.0005689
##
##
## Response selic.d :
##
## Call:
## lm(formula = selic.d ~ ect1 + ect2 + log_ipca_indice.dl1 + log_cambio.dl1 +
##     selic.dl1 + log_ipca_indice.dl2 + log_cambio.dl2 + selic.dl2 +
##     log_ipca_indice.dl3 + log_cambio.dl3 + selic.dl3 + log_ipca_indice.dl4 +
##     log_cambio.dl4 + selic.dl4 - 1, data = data.mat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.66054 -0.19978 -0.00839  0.18220  1.14967
##
## Coefficients:
##                     Estimate Std. Error t value   Pr(>|t|)
## ect1                 0.02861    0.02704   1.058    0.29108
## ect2                -0.02000    0.01292  -1.547    0.12301
## log_ipca_indice.dl1 16.45107    7.76611   2.118    0.03511 *
## log_cambio.dl1      -1.41805    0.62705  -2.261    0.02457 *
## selic.dl1            0.19450    0.06173   3.151    0.00182 **
## log_ipca_indice.dl2  6.12769    8.65276   0.708    0.47948
## log_cambio.dl2       1.52308    0.66682   2.284    0.02318 *
## selic.dl2            0.28156    0.05884   4.785 0.00000289 ***
## log_ipca_indice.dl3 12.44135    8.67853   1.434    0.15291
## log_cambio.dl3      -1.10362    0.67121  -1.644    0.10136
## selic.dl3            0.29013    0.05699   5.091 0.00000069 ***
## log_ipca_indice.dl4 -8.41328    7.83763  -1.073    0.28408
## log_cambio.dl4      -0.20335    0.62769  -0.324    0.74623
## selic.dl4           -0.15968    0.05655  -2.824    0.00512 **
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3384 on 257 degrees of freedom
## Multiple R-squared:  0.5508, Adjusted R-squared:  0.5263
## F-statistic: 22.51 on 14 and 257 DF,  p-value: < 0.00000000000000022
```

10. Pass-through por regime (VECM rank 1 imposto)

```r
## --- etapa 10: pass-through por regime (rank 1) + Portmanteau por subamostra ---
estima_pt <- function(dados, ini, fim, K = K_opt, r = 1) {
  sub <- dados[dados$data >= ini & dados$data <= fim, ]
  Ys  <- as.matrix(sub[, c("log_ipca_indice", "log_cambio", "selic")])
  Kx  <- min(K, max(2, floor(nrow(Ys) / 25)))
  jo  <- ca.jo(Ys, type = "trace", K = Kx, ecdet = "const", spec = "longrun")
  rank5 <- sum(jo@teststat > jo@cval[, "5pct"])
  fit <- cajorls(jo, r = r)
  pt  <- -fit$beta[paste0("log_cambio.l", Kx), 1]
  p_port <- serial.test(vec2var(jo, r = r), lags.pt = 12,
                        type = "PT.adjusted")$serial$p.value
  data.frame(n = nrow(Ys), K = Kx, rank_traco_5 = rank5,
             pass_through = round(pt, 3),
             portmanteau_p = round(as.numeric(p_port), 4))
}

janelas <- list(
  c("2003-01-01", "2025-12-01"),  # amostra cheia
  c("2003-01-01", "2014-12-01"),  # pré-2015
  c("2015-01-01", "2025-12-01"),  # pós-2015
  c("2003-01-01", "2019-12-01"),  # pré-COVID
  c("2020-01-01", "2025-12-01"))  # pós-COVID
nomes_reg <- c("Amostra cheia", "Pré-2015", "Pós-2015", "Pré-COVID", "Pós-COVID")

tab_sub <- do.call(rbind, lapply(janelas, function(j)
  estima_pt(df, as.Date(j[1]), as.Date(j[2]))))
tab_sub <- cbind(regime = nomes_reg,
                 janela = sapply(janelas, function(j)
                   paste0(substr(j[1], 1, 4), "-", substr(j[2], 1, 4))),
                 tab_sub)

cat("===== Pass-through de longo prazo por regime (VECM rank 1) =====\n")
```

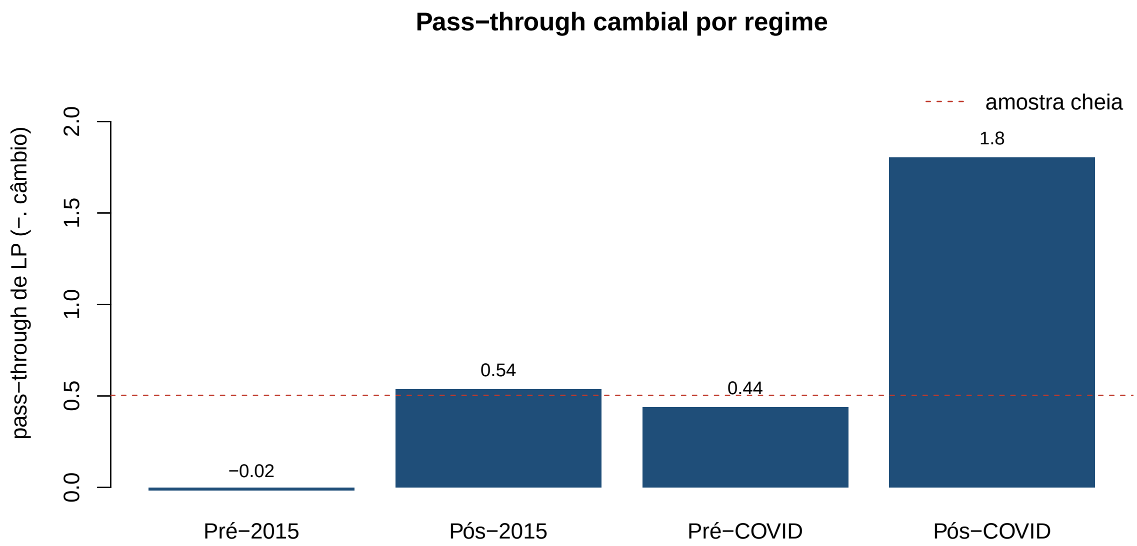
```
## ===== Pass-through de longo prazo por regime (VECM rank 1) =====
```

```r
print(tab_sub, row.names = FALSE)
```

```
##         regime    janela   n K rank_traco_5 pass_through portmanteau_p
##  Amostra cheia 2003-2025 276 5            2        0.503        0.0285
##       Pré-2015 2003-2014 144 5            2       -0.016        0.0020
##       Pós-2015 2015-2025 132 5            1        0.537        0.2083
##      Pré-COVID 2003-2019 204 5            2        0.439        0.0026
##      Pós-COVID 2020-2025  72 2            2        1.803        0.0101
```

```r
sub_reg <- tab_sub[tab_sub$regime != "Amostra cheia", ]
bp <- barplot(sub_reg$pass_through, names.arg = sub_reg$regime,
```

```r
             col = "#1f4e79", border = NA,
             ylab = "pass-through de LP (-  câmbio)",
             main = "Pass-through cambial por regime",
             ylim = range(0, sub_reg$pass_through, na.rm = TRUE) * 1.25)
abline(h = tab_sub$pass_through[tab_sub$regime == "Amostra cheia"], lty = 2, col = "#c0392b")
text(bp, sub_reg$pass_through, labels = round(sub_reg$pass_through, 2), pos = 3, cex = 0.85)
legend("topright", lty = 2, col = "#c0392b", legend = "amostra cheia", bty = "n")
```



11. VECM com 5 dummies de intervenção

```r
## --- etapa 11: VECM com dummies determinísticas (dumvar no ca.jo) ---
datas_modelo <- df$data
dum_interv <- as.matrix(data.frame(
  D_2008m10 = as.integer(format(datas_modelo, "%Y-%m") == "2008-10"),
  D_2015m09 = as.integer(format(datas_modelo, "%Y-%m") == "2015-09"),
  D_2016m01 = as.integer(format(datas_modelo, "%Y-%m") == "2016-01"),
  D_2020m03 = as.integer(format(datas_modelo, "%Y-%m") == "2020-03"),
  D_2020m08 = as.integer(format(datas_modelo, "%Y-%m") == "2020-08")
))
stopifnot(nrow(dum_interv) == nrow(Y))

jo_trace_dum <- ca.jo(Y, type = "trace", K = K_opt, ecdet = "const",
                      spec = "longrun", dumvar = dum_interv)
rank_trace_dum_5 <- sum(jo_trace_dum@teststat > jo_trace_dum@cval[, "5pct"])
rank_dum_usado   <- max(min(rank_trace_dum_5, ncol(Y) - 1), 1)

vecm_dum_as_var <- vec2var(jo_trace_dum, r = rank_dum_usado)
pt_vecm_dum     <- serial.test(vecm_dum_as_var, lags.pt = 12, type = "PT.adjusted")
norm_vecm_dum   <- normality.test(vecm_dum_as_var)

cat("Dummies: 2008-10, 2015-09, 2016-01, 2020-03, 2020-08\n")
```

```
## Dummies: 2008-10, 2015-09, 2016-01, 2020-03, 2020-08
```

```r
cat("Rank do traço a 5% (com dummies):", rank_trace_dum_5,
    "| rank usado no vec2var:", rank_dum_usado, "\n")
```

```
## Rank do traço a 5% (com dummies): 2 | rank usado no vec2var: 2
```

```r
vecm_dum_fit <- cajorls(jo_trace_dum, r = rank_dum_usado)
cat("\n===== Vetor(es) de cointegração (beta) - VECM com dummies =====\n")
```

```
##
## ===== Vetor(es) de cointegração (beta) - VECM com dummies =====
```

```r
print(round(vecm_dum_fit$beta, 4))
```

```
##                        ect1     ect2
## log_ipca_indice.l5   1.0000   0.0000
## log_cambio.l5        0.0000   1.0000
## selic.l5             1.4999   2.7026
## constant           -25.0764 -28.0090
```

```r
cat("===== summary das equações do VECM com dummies (rlm) =====\n")
```

```
## ===== summary das equações do VECM com dummies (rlm) =====
```

```r
summary(vecm_dum_fit$rlm)
```

```
## Response log_ipca_indice.d :
##
## Call:
## lm(formula = log_ipca_indice.d ~ ect1 + ect2 + D_2008m10 + D_2015m09 +
##     D_2016m01 + D_2020m03 + D_2020m08 + log_ipca_indice.dl1 +
##     log_cambio.dl1 + selic.dl1 + log_ipca_indice.dl2 + log_cambio.dl2 +
##     selic.dl2 + log_ipca_indice.dl3 + log_cambio.dl3 + selic.dl3 +
##     log_ipca_indice.dl4 + log_cambio.dl4 + selic.dl4 - 1, data = data.mat)
##
## Residuals:
##        Min         1Q     Median         3Q        Max
## -0.0109185 -0.0013164 -0.0001099  0.0014915  0.0087835
##
## Coefficients:
##                        Estimate  Std. Error t value          Pr(>|t|)
## ect1                -0.00158304  0.00023007  -6.881 0.000000000046934 ***
## ect2                 0.00086066  0.00012576   6.844 0.000000000058407 ***
## D_2008m10            0.00081440  0.00279145   0.292            0.7707
## D_2015m09            0.00183162  0.00275880   0.664            0.5073
## D_2016m01            0.00453659  0.00278064   1.631            0.1040
## D_2020m03           -0.00281418  0.00278896  -1.009            0.3139
## D_2020m08           -0.00363395  0.00286942  -1.266            0.2065
## log_ipca_indice.dl1  0.46881354  0.06199400   7.562 0.000000000000739 ***
## log_cambio.dl1       0.00233525  0.00517286   0.451            0.6521
## selic.dl1            0.00069844  0.00049337   1.416            0.1581
## log_ipca_indice.dl2  0.03130240  0.06946132   0.451            0.6526
## log_cambio.dl2       0.00106198  0.00541892   0.196            0.8448
## selic.dl2            0.00049550  0.00047275   1.048            0.2956
## log_ipca_indice.dl3 -0.02880356  0.07079471  -0.407            0.6845
## log_cambio.dl3       0.00089644  0.00536702   0.167            0.8675
## selic.dl3            0.00005899  0.00045773   0.129            0.8976
```

```
## log_ipca_indice.dl4 -0.09954065  0.06252574  -1.592            0.1126
## log_cambio.dl4       0.01243569  0.00511309   2.432            0.0157 *
## selic.dl4           -0.00027534  0.00045237  -0.609            0.5433
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.002691 on 252 degrees of freedom
## Multiple R-squared:  0.7819, Adjusted R-squared:  0.7654
## F-statistic: 47.54 on 19 and 252 DF,  p-value: < 0.00000000000000022
##
##
## Response log_cambio.d :
##
## Call:
## lm(formula = log_cambio.d ~ ect1 + ect2 + D_2008m10 + D_2015m09 +
##     D_2016m01 + D_2020m03 + D_2020m08 + log_ipca_indice.dl1 +
##     log_cambio.dl1 + selic.dl1 + log_ipca_indice.dl2 + log_cambio.dl2 +
##     selic.dl2 + log_ipca_indice.dl3 + log_cambio.dl3 + selic.dl3 +
##     log_ipca_indice.dl4 + log_cambio.dl4 + selic.dl4 - 1, data = data.mat)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.096943 -0.019309 -0.000282  0.018084  0.107124
##
## Coefficients:
##                        Estimate  Std. Error t value   Pr(>|t|)
## ect1                -0.00112790  0.00269597  -0.418    0.67604
## ect2                 0.00032318  0.00147368   0.219    0.82659
## D_2008m10            0.16231365  0.03271059   4.962 0.00000128 ***
## D_2015m09            0.08948097  0.03232799   2.768    0.00606 **
## D_2016m01            0.04656514  0.03258388   1.429    0.15422
## D_2020m03            0.10341630  0.03268143   3.164    0.00175 **
## D_2020m08            0.01938958  0.03362423   0.577    0.56469
## log_ipca_indice.dl1  0.35691079  0.72645447   0.491    0.62364
## log_cambio.dl1       0.24879350  0.06061631   4.104 0.00005479 ***
## selic.dl1           -0.00190164  0.00578136  -0.329    0.74249
## log_ipca_indice.dl2 -0.13961339  0.81395750  -0.172    0.86395
## log_cambio.dl2      -0.07733675  0.06349972  -1.218    0.22440
## selic.dl2           -0.00179104  0.00553974  -0.323    0.74673
## log_ipca_indice.dl3  0.34714367  0.82958245   0.418    0.67597
## log_cambio.dl3       0.04706211  0.06289149   0.748    0.45497
## selic.dl3            0.00304489  0.00536371   0.568    0.57076
## log_ipca_indice.dl4 -0.72998608  0.73268545  -0.996    0.32005
## log_cambio.dl4      -0.07715088  0.05991585  -1.288    0.19905
## selic.dl4            0.00006059  0.00530099   0.011    0.99089
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.03153 on 252 degrees of freedom
## Multiple R-squared:  0.2604, Adjusted R-squared:  0.2047
## F-statistic:  4.67 on 19 and 252 DF,  p-value: 0.000000003359
##
##
## Response selic.d :
```

```
##
## Call:
## lm(formula = selic.d ~ ect1 + ect2 + D_2008m10 + D_2015m09 +
##     D_2016m01 + D_2020m03 + D_2020m08 + log_ipca_indice.dl1 +
##     log_cambio.dl1 + selic.dl1 + log_ipca_indice.dl2 + log_cambio.dl2 +
##     selic.dl2 + log_ipca_indice.dl3 + log_cambio.dl3 + selic.dl3 +
##     log_ipca_indice.dl4 + log_cambio.dl4 + selic.dl4 - 1, data = data.mat)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.6801 -0.1893  0.0000  0.1828  1.1417
##
## Coefficients:
##                      Estimate Std. Error t value    Pr(>|t|)
## ect1                 0.056946   0.029064   1.959     0.05117 .
## ect2                -0.039057   0.015887  -2.458     0.01463 *
## D_2008m10           -0.117402   0.352635  -0.333     0.73947
## D_2015m09           -0.165836   0.348510  -0.476     0.63460
## D_2016m01            0.008961   0.351269   0.026     0.97967
## D_2020m03           -0.454697   0.352321  -1.291     0.19803
## D_2020m08            0.377890   0.362484   1.042     0.29818
## log_ipca_indice.dl1 15.984673   7.831508   2.041     0.04229 *
## log_cambio.dl1      -1.349144   0.653471  -2.065     0.03999 *
## selic.dl1            0.192264   0.062326   3.085     0.00226 **
## log_ipca_indice.dl2  4.685535   8.774830   0.534     0.59383
## log_cambio.dl2       1.625875   0.684556   2.375     0.01829 *
## selic.dl2            0.287454   0.059721   4.813 0.000002561 ***
## log_ipca_indice.dl3 15.712656   8.943274   1.757     0.08015 .
## log_cambio.dl3      -1.175107   0.677999  -1.733     0.08428 .
## selic.dl3            0.290216   0.057823   5.019 0.000000982 ***
## log_ipca_indice.dl4 -8.382431   7.898681  -1.061     0.28959
## log_cambio.dl4      -0.277519   0.645920  -0.430     0.66782
## selic.dl4           -0.164123   0.057147  -2.872     0.00443 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3399 on 252 degrees of freedom
## Multiple R-squared:  0.5555, Adjusted R-squared:  0.522
## F-statistic: 16.58 on 19 and 252 DF,  p-value: < 0.00000000000000022
```

```r
cat("===== Portmanteau ajustado (VECM com dummies), lags.pt = 12 =====\n")
```

```
## ===== Portmanteau ajustado (VECM com dummies), lags.pt = 12 =====
```

```r
print(pt_vecm_dum$serial)
```

```
##
##  Portmanteau Test (adjusted)
##
## data:  Residuals of VAR object vecm_dum_as_var
## Chi-squared = 102.27, df = 66, p-value = 0.002805
```

```r
cat("\n===== Jarque-Bera multivariado (VECM com dummies) =====\n")
```

```
##
## ===== Jarque-Bera multivariado (VECM com dummies) =====
```

```r
print(norm_vecm_dum$jb.mul)
```

```
## $JB
##
##  JB-Test (multivariate)
##
## data:  Residuals of VAR object vecm_dum_as_var
## Chi-squared = 173.21, df = 6, p-value < 0.00000000000000022
##
##
## $Skewness
##
##  Skewness only (multivariate)
##
## data:  Residuals of VAR object vecm_dum_as_var
## Chi-squared = 16.171, df = 3, p-value = 0.001046
##
##
## $Kurtosis
##
##  Kurtosis only (multivariate)
##
## data:  Residuals of VAR object vecm_dum_as_var
## Chi-squared = 157.04, df = 3, p-value < 0.00000000000000022
```

12. VECM com IBC-Br (atividade real)

```r
## --- etapa 12a: ordem de integração do log(IBC-Br) ---
s_ibcbr <- df$log_ibcbr

adf_ibc_trend <- ur.df(s_ibcbr,         type = "trend", selectlags = "AIC")
adf_ibc_drift <- ur.df(diff(s_ibcbr),   type = "drift", selectlags = "AIC")
kpss_ibc_n    <- ur.kpss(s_ibcbr,       type = "mu")
kpss_ibc_d    <- ur.kpss(diff(s_ibcbr), type = "mu")

cat("===== ADF log(IBC-Br) - nível (trend) =====\n");       summary(adf_ibc_trend)
```

```
## ===== ADF log(IBC-Br) - nível (trend) =====

##
## ###############################################
## # Augmented Dickey-Fuller Test Unit Root Test #
## ###############################################
##
## Test regression trend
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.083616 -0.003981  0.000081  0.005357  0.045250
```

```
##
## Coefficients:
##                Estimate  Std. Error t value Pr(>|t|)
## (Intercept)  0.10284975  0.04576655   2.247 0.025431 *
## z.lag.1     -0.02300703  0.01044885  -2.202 0.028520 *
## tt           0.00001936  0.00001458   1.327 0.185569
## z.diff.lag   0.21520838  0.05888573   3.655 0.000309 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.01124 on 270 degrees of freedom
## Multiple R-squared:  0.06398,    Adjusted R-squared:  0.05358
## F-statistic: 6.152 on 3 and 270 DF,  p-value: 0.0004645
##
##
## Value of test-statistic is: -2.2019 2.8461 2.7165
##
## Critical values for test statistics:
##       1pct  5pct 10pct
## tau3 -3.98 -3.42 -3.13
## phi2  6.15  4.71  4.05
## phi3  8.34  6.30  5.36
```

```
cat("\n===== ADF log(IBC-Br) - 1ª dif. (drift) =====\n");    summary(adf_ibc_drift)
```

```
##
## ===== ADF log(IBC-Br) - 1ª dif. (drift) =====
##
## ###############################################
## # Augmented Dickey-Fuller Test Unit Root Test #
## ###############################################
##
## Test regression drift
##
##
## Call:
## lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.080193 -0.004216  0.000256  0.005362  0.043801
##
## Coefficients:
##               Estimate Std. Error t value            Pr(>|t|)
## (Intercept)  0.0013831  0.0006917   2.000              0.0465 *
## z.lag.1     -0.8728197  0.0756339 -11.540 <0.0000000000000002 ***
## z.diff.lag   0.1133067  0.0601752   1.883              0.0608 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.01126 on 270 degrees of freedom
## Multiple R-squared:  0.4001, Adjusted R-squared:  0.3956
## F-statistic: 90.02 on 2 and 270 DF,  p-value: < 0.00000000000000022
##
```

```
##
## Value of test-statistic is: -11.5401 66.5866
##
## Critical values for test statistics:
##       1pct  5pct 10pct
## tau2 -3.44 -2.87 -2.57
## phi1  6.47  4.61  3.79
```

```r
cat("\n===== KPSS log(IBC-Br) - nível =====\n");            summary(kpss_ibc_n)
```

```
##
## ===== KPSS log(IBC-Br) - nível =====

##
## #######################
## # KPSS Unit Root Test #
## #######################
##
## Test is of type: mu with 5 lags.
##
## Value of test-statistic is: 3.1094
##
## Critical value for a significance level of:
##                 10pct  5pct 2.5pct  1pct
## critical values 0.347 0.463  0.574 0.739
```

```r
cat("\n===== KPSS log(IBC-Br) - 1ª dif. =====\n");          summary(kpss_ibc_d)
```

```
##
## ===== KPSS log(IBC-Br) - 1ª dif. =====

##
## #######################
## # KPSS Unit Root Test #
## #######################
##
## Test is of type: mu with 5 lags.
##
## Value of test-statistic is: 0.224
##
## Critical value for a significance level of:
##                 10pct  5pct 2.5pct  1pct
## critical values 0.347 0.463  0.574 0.739
```

```r
## --- etapa 12b: sistema de 4 variáveis (IBC-Br endógeno) ---
Y_ibc <- data.frame(
  log_ipca_indice = s_ipca,
  log_cambio      = s_cambio,
  selic           = s_selic,
  log_ibcbr       = s_ibcbr
)

var_sel_ibc <- VARselect(Y_ibc, lag.max = 12, type = "const")
K_ibc       <- max(as.integer(var_sel_ibc$selection["AIC(n)"]), 2)
cat("Defasagem AIC (sistema 4 variáveis): K =", K_ibc, "\n\n")
```

```
## Defasagem AIC (sistema 4 variáveis): K = 4
```

```r
print(var_sel_ibc$selection)
```

```
## AIC(n)  HQ(n)  SC(n) FPE(n)
##      4      4      2      4
```

```r
jo_ibc         <- ca.jo(Y_ibc, type = "trace", K = K_ibc, ecdet = "const", spec = "longrun")
rank_ibc       <- sum(jo_ibc@teststat > jo_ibc@cval[, "5pct"])
rank_ibc_usado <- max(min(rank_ibc, ncol(Y_ibc) - 1), 1)

cat("\nRank do traço a 5% (com IBC-Br):", rank_ibc,
    "| rank usado:", rank_ibc_usado, "\n")
```

```
##
## Rank do traço a 5% (com IBC-Br): 2 | rank usado: 2
```

```r
cat("===== Johansen por traço - sistema 4 variáveis (com IBC-Br) =====\n")
```

```
## ===== Johansen por traço - sistema 4 variáveis (com IBC-Br) =====
```

```r
summary(jo_ibc)
```

```
##
## ######################
## # Johansen-Procedure #
## ######################
##
## Test type: trace statistic , without linear trend and constant in cointegration
##
## Eigenvalues (lambda):
## [1]  0.20912140829364034289028  0.10644733438945876147219
## [3]  0.04630941368457520274093  0.01877992293858585584609
## [5] -0.0000000000000000004553766
##
## Values of teststatistic and critical values of test:
##
##            test 10pct  5pct  1pct
## r <= 3 |   5.16  7.52  9.24 12.97
## r <= 2 |  18.05 17.85 19.96 24.60
## r <= 1 |  48.67 32.00 34.91 41.07
## r = 0  | 112.48 49.65 53.12 60.16
##
## Eigenvectors, normalised to first column:
## (These are the cointegration relations)
##
##                    log_ipca_indice.l4 log_cambio.l4     selic.l4 log_ibcbr.l4
## log_ipca_indice.l4         1.00000000    1.00000000  1.000000000    1.0000000
## log_cambio.l4             -0.55287937   -0.55119992 -0.584844069    0.4434262
## selic.l4                  -0.00922752    0.01799529 -0.007306859   -0.1247922
## log_ibcbr.l4              -1.31741283   -1.99366711 -2.011794650  -13.1703769
## constant                  -2.39821360    1.10679550  1.556821427   52.2701065
##                       constant
## log_ipca_indice.l4  1.00000000
## log_cambio.l4       0.44580134
## selic.l4            0.01972129
## log_ibcbr.l4       -0.94240674
## constant           -4.70543469
```

```
##
## Weights W:
## (This is the loading matrix)
##
##                    log_ipca_indice.l4 log_cambio.l4     selic.l4   log_ibcbr.l4
## log_ipca_indice.d       -0.003055690 -0.0065035058 -0.001784595 -0.00002767276
## log_cambio.d             0.001019933 -0.0075520231  0.071182367 -0.00345615764
## selic.d                  0.238135937 -1.1838183576  0.136764367  0.01050138886
## log_ibcbr.d             -0.005884287  0.0008556205  0.016180500  0.00126375406
##                                     constant
## log_ipca_indice.d  0.000000000000000001803397
## log_cambio.d      -0.000000000000000021481446
## selic.d            0.000000000000000085780173
## log_ibcbr.d        0.000000000000000002425410
```

```
vecm_ibc_as_var <- vec2var(jo_ibc, r = rank_ibc_usado)
pt_vecm_ibc     <- serial.test(vecm_ibc_as_var, lags.pt = 12, type = "PT.adjusted")

# pass-through de longo prazo: beta_1 normalizado em log(IPCA)
pt_coef <- function(jo, j = 1) {
  b  <- jo@V[, j]; rn <- rownames(jo@V)
  bi <- b[grep("^log_ipca",   rn)[1]]
  bc <- b[grep("^log_cambio", rn)[1]]
  as.numeric(-(bc / bi))
}
passthrough_ibc <- pt_coef(jo_ibc, 1)

vecm_ibc_fit <- cajorls(jo_ibc, r = rank_ibc_usado)
cat("===== Vetor(es) de cointegração (beta) - VECM com IBC-Br =====\n")
```

```
## ===== Vetor(es) de cointegração (beta) - VECM com IBC-Br =====
```

```
print(round(vecm_ibc_fit$beta, 4))
```

```
##                         ect1      ect2
## log_ipca_indice.l4    1.0000    0.0000
## log_cambio.l4         0.0000    1.0000
## selic.l4              8.9526   16.2093
## log_ibcbr.l4       -223.9416 -402.6632
## constant           1151.4572 2086.9931
```

```
cat("\nPass-through de longo prazo (normalizado em log IPCA):",
    round(passthrough_ibc, 3), "\n")
```

```
##
## Pass-through de longo prazo (normalizado em log IPCA): 0.553
```

```
cat("\n===== Portmanteau ajustado (VECM + IBC-Br), lags.pt = 12 =====\n")
```

```
##
## ===== Portmanteau ajustado (VECM + IBC-Br), lags.pt = 12 =====
```

```
print(pt_vecm_ibc$serial)
```

```
##
##  Portmanteau Test (adjusted)
##
## data:  Residuals of VAR object vecm_ibc_as_var
```

```
## Chi-squared = 177.14, df = 132, p-value = 0.005345
```

```r
## --- etapa 12c: VECM + IBC-Br SEM Selic ---
Y_sem <- data.frame(
  log_ipca_indice = s_ipca,
  log_cambio      = s_cambio,
  log_ibcbr       = s_ibcbr
)
var_sel_sem    <- VARselect(Y_sem, lag.max = 12, type = "const")
K_sem          <- max(as.integer(var_sel_sem$selection["AIC(n)"]), 2)
jo_sem         <- ca.jo(Y_sem, type = "trace", K = K_sem, ecdet = "const", spec = "longrun")
rank_sem       <- sum(jo_sem@teststat > jo_sem@cval[, "5pct"])
rank_sem_usado <- max(min(rank_sem, ncol(Y_sem) - 1), 1)
vecm_sem_as_var <- vec2var(jo_sem, r = rank_sem_usado)
pt_vecm_sem     <- serial.test(vecm_sem_as_var, lags.pt = 12, type = "PT.adjusted")
passthrough_sem <- pt_coef(jo_sem, 1)

cat("Defasagem AIC (sem Selic): K =", K_sem,
    "| rank do traço a 5%:", rank_sem, "| rank usado:", rank_sem_usado, "\n")
```

```
## Defasagem AIC (sem Selic): K = 2 | rank do traço a 5%: 3 | rank usado: 2
```

```r
cat("Pass-through (sem Selic):", round(passthrough_sem, 3), "\n\n")
```

```
## Pass-through (sem Selic): 0.553
```

```r
cat("===== Portmanteau ajustado (VECM + IBC-Br, sem Selic), lags.pt = 12 =====\n")
```

```
## ===== Portmanteau ajustado (VECM + IBC-Br, sem Selic), lags.pt = 12 =====
```

```r
print(pt_vecm_sem$serial)
```

```
##
##  Portmanteau Test (adjusted)
##
## data:  Residuals of VAR object vecm_sem_as_var
## Chi-squared = 117.01, df = 93, p-value = 0.04684
```

12.1 Escada consolidada de Portmanteau

```r
## --- etapa 12d: escada completa de Portmanteau (lags.pt = 12) ---
extrair_stat <- function(x) as.numeric(x$statistic)
extrair_pval <- function(x) as.numeric(x$p.value)

escada <- data.frame(
  especificacao = c("VECM-base", "VECM com K = 6",
                    "VECM + dummies de intervenção",
                    "VECM + IBC-Br (atividade real)",
                    "VECM + IBC-Br, sem Selic"),
  k    = c(3, 3, 3, 4, 3),
  rank = c(rank_coint, rank_coint, rank_dum_usado, rank_ibc_usado, rank_sem_usado),
  portmanteau = round(c(extrair_stat(pt_vecm$serial),  extrair_stat(pt_vecm6$serial),
                        extrair_stat(pt_vecm_dum$serial), extrair_stat(pt_vecm_ibc$serial),
                        extrair_stat(pt_vecm_sem$serial)), 2),
  p_valor = round(c(extrair_pval(pt_vecm$serial),  extrair_pval(pt_vecm6$serial),
                    extrair_pval(pt_vecm_dum$serial), extrair_pval(pt_vecm_ibc$serial),
```

```r
                    extrair_pval(pt_vecm_sem$serial)), 4)
)
escada$diagnostico <- ifelse(escada$p_valor < 0.05,
                             "rejeita ruido branco", "nao rejeita ruido branco")
cat("===== Escada de Portmanteau ajustado (lags.pt = 12) por especificação =====\n")
```

```
## ===== Escada de Portmanteau ajustado (lags.pt = 12) por especificação =====
```

```r
print(escada, row.names = FALSE)
```

```
##                    especificacao k rank portmanteau p_valor          diagnostico
##                        VECM-base 3    2       90.20  0.0256 rejeita ruido branco
##                   VECM com K = 6 3    2       89.76  0.0037 rejeita ruido branco
##    VECM + dummies de intervenção 3    2      102.27  0.0028 rejeita ruido branco
##   VECM + IBC-Br (atividade real) 4    2      177.14  0.0053 rejeita ruido branco
##         VECM + IBC-Br, sem Selic 3    2      117.01  0.0468 rejeita ruido branco
```

13. Conferência dos números-chave

```r
## --- etapa 13: confirmação dos números que devem bater com o PDF final ---
# Pass-through de longo prazo sob rank 1 (vetor único, IPCA = 1)
vecm_r1 <- cajorls(jo_trace, r = 1)
nm_cambio <- paste0("log_cambio.l", K_opt)
pt_r1 <- -vecm_r1$beta[nm_cambio, 1]

cat("--------------------------------------------------------------\n")
```

```
## --------------------------------------------------------------
```

```r
cat("CONFERÊNCIA DE NÚMEROS-CHAVE (devem bater com o PDF final)\n")
```

```
## CONFERÊNCIA DE NÚMEROS-CHAVE (devem bater com o PDF final)
```

```r
cat("--------------------------------------------------------------\n")
```

```
## --------------------------------------------------------------
```

```r
cat(sprintf("Pass-through (rank 1, -beta câmbio)  : %.4f  [esperado ~0,503]\n", pt_r1))
```

```
## Pass-through (rank 1, -beta câmbio)  : 0.5031  [esperado ~0,503]
```

```r
ts_trace <- jo_trace@teststat
names(ts_trace) <- rownames(jo_trace@cval)
cat(sprintf("Johansen traço r=0                   : %.2f  [esperado 75,87]\n",
            ts_trace["r = 0"]))
```

```
## Johansen traço r=0                   : NA  [esperado 75,87]
```

```r
cat(sprintf("Johansen traço r<=1                  : %.2f  [esperado 22,67]\n",
            ts_trace["r <= 1"]))
```

```
## Johansen traço r<=1                  : NA  [esperado 22,67]
```

```r
cat(sprintf("Johansen traço r<=2                  : %.2f  [esperado 5,56]\n",
            ts_trace["r <= 2"]))
```

```
## Johansen traço r<=2                  : NA  [esperado 5,56]
```

```r
cat(sprintf("Portmanteau VECM-base (p-valor)      : %.4f  [esperado 0,0256]\n",
            as.numeric(pt_vecm$serial$p.value)))
```

```
## Portmanteau VECM-base (p-valor)      : 0.0256  [esperado 0,0256]
```

```r
cat("---------------------------------------------------------------\n")
```

```
## ---------------------------------------------------------------
```